

A Survey on Collaborative Mechanisms Between Large and Small Language Models

Yi Chen^{1,2}, JiaHao Zhao^{1,2}, HaoHao Han^{1,2}

¹Chengdu Institute of Computer Applications, Chinese Academy of Sciences, Chengdu, China

²University of Chinese Academy of Sciences, Beijing, China

{chenyi24,zhaojiahao241,hanhaohao24}@mailsucas.ac.cn

Abstract

Large Language Models (LLMs) deliver powerful AI capabilities but face deployment challenges due to high resource costs and latency, whereas Small Language Models (SLMs) offer efficiency and deployability at the cost of reduced performance. Collaboration between LLMs and SLMs emerges as a crucial paradigm to synergistically balance these trade-offs, enabling advanced AI applications, especially on resource-constrained edge devices. This survey provides a comprehensive overview of LLM-SLM collaboration, detailing various interaction mechanisms (pipeline, routing, auxiliary, distillation, fusion), key enabling technologies, and diverse application scenarios driven by on-device needs like low latency, privacy, personalization, and offline operation. While highlighting the significant potential for creating more efficient, adaptable, and accessible AI, we also discuss persistent challenges including system overhead, inter-model consistency, robust task allocation, evaluation complexity, and security/privacy concerns. Future directions point towards more intelligent adaptive frameworks, deeper model fusion, and expansion into multimodal and embodied AI, positioning LLM-SLM collaboration as a key driver for the next generation of practical and ubiquitous artificial intelligence.

1 Introduction

1.1 Research Background and Motivation

In recent years, Large Language Models (LLMs) have achieved breakthrough progress in fields such as natural language processing, code generation, and intelligent question answering (**brown2020language**; **openai2023gpt4**). However, as the scale of model parameters continues to grow, their consumption of computing resources, energy costs, and deployment expenses have also risen significantly. Particularly in edge scenarios requiring low latency and high privacy, such as smartphones, IoT devices, and edge servers, the traditional cloud-based LLM inference model faces serious feasibility challenges (**zhou2024benchmark**). Concurrently, Small Language Models (SLMs) are widely used in resource-constrained devices due to their lightweight structure, fast inference speed, and ease of deployment (**gao2025strategic**). Against this backdrop, the collaborative mechanism between large and small language models has gradually become an important research direction in both industry and academia. It aims to leverage the powerful capabilities of LLMs and the high efficiency of SLMs through intelligent collaboration, building a more intelligent, efficient, and reliable inference system.

大语言模型与小语言模型协同机制综述

陈毅^{1,2}, 赵佳豪^{1,2}, 韩浩浩^{1,2}

¹ 中国科学院成都计算机应用研究所, 成都, 中国

² 中国科学院大学, 北京, 中国

{chenyi24,zhaojiahao241,hanhaohao24}@mailsucas.ac.cn

摘要

大语言模型 (LLMs) 提供了强大的人工智能能力, 但由于高昂的资源成本和延迟问题而面临部署挑战; 而小语言模型 (SLMs) 则提供了高效性和可部署性, 但性能有所降低。大语言模型与小语言模型之间的协同成为一个关键的范式, 旨在协同平衡这些权衡, 从而推动先进人工智能应用的发展, 尤其是在资源受限的边缘设备上。本综述全面概述了 LLM-SLM 协同, 详细阐述了各种交互机制 (流水线、路由、辅助、蒸馏、融合)、关键使能技术, 以及由低延迟、隐私、个性化和离线操作等设备端需求驱动的多样化应用场景。在强调创建更高效、更适应、更易访问的人工智能的巨大潜力的同时, 我们也讨论了持续存在的挑战, 包括系统开销、模型间一致性、鲁棒的任务分配、评估复杂性以及安全/隐私问题。未来的方向指向更智能的自适应框架、更深度的模型融合, 以及向多模态和具身人工智能的扩展, 这使 LLM-SLM 协同成为下一代实用且无处不在的人工智能的关键驱动力。

1 引言

1.1 研究背景与动机

近年来, 大语言模型在自然语言处理、代码生成、智能问答等领域取得了突破性进展 (**brown2020language**; **openai2023gpt4**)。然而, 随着模型参数规模的持续增长, 其计算资源消耗、能源成本以及部署开销也显著上升。特别是在需要低延迟和高隐私的边缘场景, 如智能手机、物联网设备和边缘服务器中, 传统基于云端的大语言模型推理模式面临着严重的可行性挑战 (**zhou2024benchmark**)。与此同时, 小语言模型因其轻量化的结构、快速的推理速度和易于部署的特点, 在资源受限的设备上得到了广泛应用 (**gao2025strategic**)。在此背景下, 大语言模型与小语言模型之间的协同机制逐渐成为工业界和学术界的重要研究方向。其旨在通过智能协作, 利用大语言模型的强大能力与小语言模型的高效性, 构建一个更智能、更高效、更可靠的推理系统。

1.2 Definition and Scope of Large-Small Model Collaboration

Broadly speaking, large-small model collaboration refers to a mechanism where LLMs and SLMs work together within a system, complementing each other's strengths. This paradigm can be divided into several sub-directions, such as pipeline collaboration, parallel collaboration, conditional triggered inference, and knowledge distillation (**wang2024comprehensive**; **gao2025strategic**). Pipeline collaboration is a sequential execution mode where the output of one model serves as the input for another (**wang2024comprehensive**). Since SLMs typically offer higher efficiency, they are often used for preliminary processing or generating candidate results, which are then passed to LLMs for more complex reasoning or knowledge integration (**gao2025strategic**). In the Cascade Speculative Decoding framework, an SLM generates draft responses, while an LLM performs verification and correction in parallel, thereby improving response speed (**chen2023speculative**). In terms of conditional inference, research has proposed using confidence scores to determine whether to invoke an LLM, thus achieving on-demand activation (**gupta2023adaptive**). Furthermore, knowledge distillation allows the knowledge from an LLM to be compressed and transferred to an SLM through training, enhancing the latter's ability to model complex tasks while retaining its computational advantages (**gu2024minillm**). These studies constitute the core scope of large-small model collaboration and provide a theoretical and practical foundation for building energy-efficient AI systems.

1.3 The Rise of On-Device Large Models and Its Impetus for Collaboration Research

The emergence of on-device large language models is propelling research into collaborative mechanisms to a new stage. In recent years, with improvements in chip performance and the maturation of model compression techniques, several tech giants have begun deploying proprietary large models on terminals. In 2024, Apple integrated an on-device language model with approximately 3 billion parameters into its "Apple Intelligence" system to handle some natural language tasks locally, while complex requests are delegated to cloud-based models (**apple2024ai**). Similarly, Huawei introduced a lightweight version of its Pangu large model into its HarmonyOS intelligent assistant, enabling hybrid edge-cloud inference (**huawei2024pangu**). This architecture prompts researchers to delve deeper into questions such as: How can the task boundaries between LLMs and SLMs be allocated more effectively? How can computation offloading and dynamic routing be managed between the edge and the cloud? How can collaboration mechanisms improve overall energy efficiency and response quality? Related challenges include model selection strategies, edge-cloud communication overhead, privacy protection mechanisms, and multi-model fusion accuracy (**zhang2025efficient**). Therefore, studying the collaboration mechanisms of large and small models is not only a hot topic for theoretical exploration but also a key technological path for intelligent terminals moving towards localized, high-performance, and low-power AI inference.

1.4 Structure and Contributions of This Paper

This paper first provides an introduction, outlining the research background, motivation, and the definition and scope of large-small model collaboration. Next, it elaborates on related concepts and foundations, including the characteristics, advantages, and limitations of large and small language models, as well as the basis and necessity for their collaboration. The third section delves into the mechanisms and architectures of large-small model collaboration, classifying and introducing modes such as pipeline, hybrid/routing, auxiliary/enhancement, knowledge distillation-driven, and integration/fusion, and analyzes the key technologies for

1.2 大小模型协作的定义与范畴

广义而言,大小模型协作是指在一个系统中,大语言模型与小语言模型协同工作、优势互补的机制。这一范式可细分为多个子方向,例如流水线协作、并行协作、条件触发式推理以及知识蒸馏等(**wang2024comprehensive**; **gao2025strategic**)。流水线协作是一种顺序执行模式,其中一个模型的输出作为另一个模型的输入(**wang2024comprehensive**)。由于小语言模型通常具有更高的效率,它们常被用于初步处理或生成候选结果,然后传递给大语言模型进行更复杂的推理或知识整合(**gao2025strategic**)。在级联推测解码框架中,小语言模型生成草稿响应,而大语言模型并行进行验证和修正,从而提升响应速度(**chen2023speculative**)。在条件推理方面,研究提出了利用置信度分数来决定是否调用大语言模型,从而实现按需激活(**gupta2023adaptive**)。此外,知识蒸馏使得大语言模型的知识能够通过训练被压缩并迁移到小语言模型中,在保留其计算优势的同时,增强后者对复杂任务的建模能力(**gu2024minillm**)。这些研究构成了大小模型协作的核心范畴,并为构建高效的人工智能系统提供了理论与实践基础。

1.3 端侧大模型的兴起及其对协作研究的推动

端侧大语言模型的出现,正将协作机制研究推向新阶段。近年来,随着芯片性能提升和模型压缩技术成熟,多家科技巨头已开始终端部署自研大模型。2024年,苹果将约30亿参数的端侧语言模型集成至“Apple Intelligence”系统,以本地处理部分自然语言任务,复杂请求则交由云端模型处理(**apple2024ai**)。同样,华为也将轻量化盘古大模型引入鸿蒙智能助手,实现边云协同推理(**huawei2024pangu**)。这种架构促使研究者更深入探讨:如何更有效划分LLM与SLM的任务边界?如何管理边云间的计算卸载与动态路由?协作机制如何提升整体能效与响应质量?相关挑战包括模型选择策略、边云通信开销、隐私保护机制及多模型融合精度等(**zhang2025efficient**)。因此,研究大小模型协作机制不仅是理论探索的热点,更是智能终端迈向本地化、高性能、低功耗AI推理的关键技术路径。

1.4 本文结构与贡献

本文首先给出引言,阐述研究背景、动机以及大小模型协作的定义与范畴。接着详述相关概念与基础,包括大语言模型与小语言模型的特点、优势与局限,以及二者协作的基础与必要性。第三部分深入探讨大小模型协作的机制与架构,对流水线式、混合/路由式、辅助/增强式、知识蒸馏驱动式、集成/融合式等模式进行分类介绍,并分析实现协作的关键技术。随后,在端侧需求驱动下,讨论协作机制的应用场景,按实时低延迟推理、隐私敏感、任务特定定制、离线/弱网环境、能耗约束等类别展开。最后,提出当前挑战与开放问题,展望未来发展趋势,并总结全文研究意义、价值与思考。

2 相关概念与基础

在深入探讨大语言模型(LLMs)与小语言模型(SLMs)之间的协作机制之前,有必要系统回顾这两类模型的基本概念、结构特点、优势与局限。它们在模型能力、资源消耗、部署场

implementing collaboration. Subsequently, driven by on-device requirements, it discusses the application scenarios of collaborative mechanisms, categorized by real-time low-latency inference, privacy sensitivity, task-specific customization, offline/weak network environments, and energy constraints. Finally, it presents current challenges and open questions, looks ahead to future development trends, and summarizes the research significance, value, and reflections of the entire paper.

2 Related Concepts and Foundations

Before delving into the collaborative mechanisms between Large Language Models (LLMs) and Small Language Models (SLMs), it is necessary to systematically review the basic concepts, structural characteristics, advantages, and limitations of these two types of models. They exhibit significant differences in model capabilities, resource consumption, deployment scenarios, and research paths. It is precisely this heterogeneity that forms the fundamental motivation for research on model collaboration.

2.1 Large Language Models

2.1.1 Definition, Architecture, and Characteristics

Large Language Models (LLMs) typically refer to language understanding and generation models with parameter scales in the billions or more. Their architectures are mostly based on the Transformer framework, employing autoregressive (e.g., GPT series) or masked language modeling strategies (e.g., BERT, T5) for pre-training (**brown2020language**; **raffel2020exploring**; **openai2023gpt4**). These models learn language structure, knowledge representation, and reasoning patterns from massive unsupervised corpora, enabling deep modeling of natural language text and supporting generalization to various downstream tasks.

From an architectural perspective, LLMs usually consist of tens to hundreds of Transformer layers, each containing multi-head attention mechanisms, feed-forward networks, residual connections, and normalization operations. As the number of model parameters increases, their capabilities do not grow linearly but exhibit certain "emergent phenomena." That is, once the model scale exceeds a certain threshold, they automatically demonstrate more complex abilities such as language understanding, mathematical reasoning, and multimodal interaction (**wei2022emergent**; **gao2025strategic**). This non-linear performance leap has become an important clue in the current exploration of artificial general intelligence.

2.1.2 Advantages and Limitations

LLMs possess unparalleled advantages in general language modeling. They exhibit strong generalization capabilities and can perform various natural language processing tasks through prompts or in-context learning, including question answering, translation, summarization, code generation, and knowledge retrieval, demonstrating potential for zero-shot and few-shot learning. LLMs have deep semantic understanding and long-text modeling capabilities, preserving contextual logic and structural integrity. Through fine-tuning or reinforcement learning (such as RLHF), their behavior and alignment with human preferences can be further optimized.

However, the limitations of LLMs cannot be ignored. Firstly, resource consumption is extremely high. Training often requires high-performance computing clusters (such as GPUs/TPUs) and weeks to months of time cost, while the inference stage also faces issues like large memory footprint and high latency. Their large size makes them difficult to deploy on resource-constrained terminal or edge devices, limiting their real-time application scenarios. Due to the uncontrollable nature of data during pre-training, models may exhibit factual errors, bias

景及研究路径上均存在显著差异。正是这种异质性构成了模型协作研究的基本动因。

2.1 大语言模型

2.1.1 定义、架构与特性

大语言模型（LLMs）通常指参数数量达到数十亿或以上的语言理解与生成模型。其架构大多基于 Transformer 框架，采用自回归（如 GPT 系列）或掩码语言建模策略（如 BERT、T5）进行预训练 (**brown2020language**; **raffel2020exploring**; **openai2023gpt4**)。这些模型从海量无监督语料中学习语言结构、知识表示和推理模式，能够对自然语言文本进行深度建模，并支持泛化到多种下游任务。

从架构角度看，LLMs 通常由数十至数百个 Transformer 层堆叠而成，每层包含多头注意力机制、前馈网络、残差连接和归一化操作。随着模型参数数量的增加，其能力并非线性增长，而是呈现出一定的“涌现现象”。即当模型规模超过某个阈值后，它们会自动展现出更复杂的语言理解、数学推理和多模态交互等能力 (**wei2022emergent**; **gao2025strategic**)。这种非线性的性能跃迁已成为当前探索通用人工智能的重要线索。

2.1.2 优势与局限

LLMs 在通用语言建模方面具有无可比拟的优势。它们展现出强大的泛化能力，能够通过提示或上下文学习完成多种自然语言处理任务，包括问答、翻译、摘要、代码生成和知识检索，显示出零样本和少样本学习的潜力。LLMs 具备深度的语义理解和长文本建模能力，能够保持上下文逻辑和结构完整性。通过微调或强化学习（如 RLHF），可以进一步优化其行为表现和与人类偏好的对齐。

然而，LLMs 的局限性也不容忽视。首先，资源消耗极高。训练通常需要高性能计算集群（如 GPU/TPU）和数周至数月的时间成本，而推理阶段也面临内存占用大、延迟高等问题。其庞大的体积使其难以部署在资源受限的终端或边缘设备上，限制了实时应用场景。由于预训练阶段数据的不可控性，模型可能出现事实性错误、偏见放大和幻觉风险，增加了实际应用中的安全与伦理成本 (**zhang2025efficient**)。LLMs 的闭源部署以及依赖云平台进行大模型调用也引发了数据隐私与合规性的担忧 (**zhou2024benchmark**)。

2.2 小语言模型

2.2.1 定义与特征

小语言模型（SLMs）指参数数量级在数百万至数亿之间的模型。其设计目标是在语言建模能力与部署效率之间取得平衡，在保证基本语言理解能力的同时，显著降低计算开销与硬件需求。结构上，SLMs 通常是大型模型的压缩版本，采用更少的 Transformer 层、更小的隐藏维度与更少的注意力头，同时在推理路径上尽可能简化计算流程 (**gu2024minillm**; **apple2024ai**)。

与 LLMs 相比，SLMs 更适合在计算资源受限的平台部署，例如边缘设备、移动终端与浏览器插件等。其具备推理速度快、能耗低、部署灵活等特点，是实现本地化、低延迟人机交互

amplification, and hallucination risks, increasing the safety and ethical costs in practical applications (**zhang2025efficient**). The closed-source deployment of LLMs and the reliance on cloud platforms for large model calls also raise concerns about data privacy and compliance (**zhou2024benchmark**).

2.2 Small Language Models

2.2.1 Definition and Characteristics

Small Language Models (SLMs) refer to models with parameter scales ranging from millions to a few hundred million. Their design goal is to balance language modeling capabilities with deployment efficiency, significantly reducing computational overhead and hardware requirements while ensuring basic language understanding capabilities. Structurally, SLMs are often compressed versions of large models, employing fewer Transformer layers, smaller hidden dimensions, and fewer attention heads, while streamlining the computation flow in the inference path as much as possible (**gu2024minillm**; **apple2024ai**).

Compared to LLMs, SLMs are more suitable for deployment on platforms with limited computing resources, such as edge devices, mobile terminals, and browser plugins. Their characteristics, including fast inference speed, low energy consumption, and flexible deployment, make them key components for achieving localized, low-latency human-computer interaction. In application scenarios like voice assistants, text input methods, and intelligent customer service, small models can respond quickly to user requests, protect user privacy, and reduce reliance on networks and the cloud.

2.2.2 Advantages, Limitations, and Typical Construction Methods

The prominent advantages of small models are reflected in their lightweight nature, high efficiency, and deployment-friendliness. On the one hand, due to their compact structure, they can perform inference on ordinary CPUs or low-power devices without relying on high-performance GPUs, significantly reducing operational costs. On the other hand, SLMs typically possess good task adaptability and can achieve performance optimization in specific scenarios through fine-tuning or distillation. Furthermore, keeping data processing local helps enhance user privacy protection capabilities, meeting the requirements of certain industries with high compliance demands.

However, it is undeniable that SLMs still have significant gaps compared to LLMs in terms of knowledge scale, semantic expression, and reasoning depth. Due to limited model capacity, they struggle to cover knowledge across a wide range of domains and often exhibit insufficient understanding, vague generation, or logical incoherence when facing open-domain tasks. In handling tasks such as long text processing, complex reasoning, and multi-turn dialogues, small models often find it difficult to maintain semantic coherence and information consistency. Additionally, their generalization ability is weaker, performing less well than LLMs in cross-domain transfer or zero-shot learning (**gupta2023adaptive**).

Currently, the mainstream construction methods for SLMs can be categorized into three types: First, knowledge distillation, which compresses the capabilities of large models into small models using a teacher-student framework, such as DistilGPT and TinyLLaMA; second, building lightweight models from scratch through independent training on medium-scale corpora, such as MiniBERT and ALBERT; third, pruning, quantization, or low-rank decomposition based on existing model architectures to compress size while retaining some original performance (**wang2024comprehensive**). These methods collectively promote the widespread application of small models in industrial deployment and provide a technical reserve for subsequent research on collaborative mechanisms.

的关键组成部分。在语音助手、文本输入法、智能客服等应用场景中，小模型能够快速响应用户请求，保护用户隐私，并降低对网络与云端的依赖。

2.2.2 优势、局限性与典型构建方法

小模型的突出优势体现在其轻量化、高效率与部署友好性上。一方面，由于其结构紧凑，可在普通 CPU 或低功耗设备上进行推理，无需依赖高性能 GPU，显著降低了运营成本。另一方面，SLMs 通常具备较好的任务适应性，可通过微调或蒸馏等方式在特定场景下实现性能优化。此外，保持数据处理本地化有助于提升用户隐私保护能力，满足某些合规要求较高的行业需求。

然而不可否认，SLMs 在知识规模、语义表达与推理深度等方面与 LLMs 仍存在明显差距。由于模型容量有限，其难以覆盖广泛领域的知识，面对开放域任务时常常表现出理解不足、生成模糊或逻辑不连贯等问题。在处理长文本处理、复杂推理与多轮对话等任务时，小模型往往难以维持语义连贯性与信息一致性。此外，其泛化能力较弱，在跨领域迁移或零样本学习方面表现不如 LLMs (**gupta2023adaptive**)。

目前，SLMs 的主流构建方法可归纳为三类：第一，知识蒸馏，通过师生框架将大模型能力压缩至小模型，如 DistilGPT 与 TinyLLaMA；第二，通过在中规模语料上独立训练从头构建轻量模型，如 MiniBERT 与 ALBERT；第三，基于现有模型架构进行剪枝、量化或低秩分解，在压缩尺寸的同时保留部分原有性能 (**wang2024comprehensive**)。这些方法共同推动了小模型在工业部署中的广泛应用，并为后续协同机制的研究提供了技术储备。

2.3 大语言模型与小语言模型协同的基础与必要性

2.3.1 协同的技术基础：能力与特征的互补性

大语言模型（LLMs）与小语言模型（SLMs）在能力、架构与部署特征上存在显著差异，构成了协同的自然技术基础。

2.3.2 能力边界的互补性

大模型的优势包括强大的上下文理解能力、逻辑推理能力以及多任务泛化能力（例如，GPT-4 在数学推理、代码生成等复杂任务上准确率超过 90%），但其依赖海量训练数据（万亿级参数、千亿级语料）与高算力支持（单次推理成本约 0.1–1）。

反之，小模型的优势在于参数量较小（通常低于 10 亿参数，如 TinyBERT、DistilBERT）、部署灵活（可在手机、智能音箱等边缘设备运行）以及推理速度快（延迟低于 10ms，为大模型的 1/10）。但其在复杂逻辑与长上下文处理方面能力受限（例如，数学题准确率不足 50%）。

协作本质上是“大模型”充当复杂决策的“大脑”，小模型作为轻量交互的“神经末梢”，形成“中央处理器 + 边缘节点”的分工体系。

2.3 Foundation and Necessity of Collaboration Between Large and Small Language Models

2.3.1 Technical Foundation for Collaboration: Complementarity of Capabilities and Characteristics

Large Language Models (LLMs) and Small Language Models (SLMs) exhibit significant differences in capabilities, architecture, and deployment characteristics, forming a natural technical foundation for collaboration.

2.3.2 Complementarity of Capability Boundaries

The advantages of large models include strong contextual understanding, logical reasoning, and multi-task generalization capabilities (e.g., GPT-4 achieves over 90% accuracy on complex tasks like mathematical reasoning and code generation), but they rely on massive training data (trillion-level parameters, hundred-billion-level corpora) and high computational power support (single inference cost is about 0.1–1).

Conversely, the advantages of small models lie in their smaller parameter count (typically under 1 billion parameters, e.g., TinyBERT, DistilBERT), flexible deployment (can run on edge devices like phones, smart speakers), and fast inference speed (latency below 10ms, 1/10th that of large models). However, their capabilities are limited in complex logic and long-context processing (e.g., accuracy on math problems is less than 50%).

Collaboration essentially involves the large model acting as the "brain" for complex decision-making, while the small model serves as the "nerve endings" for lightweight interaction, forming a "central processor + edge node" division of labor system.

2.3.3 Transferability of Architecture and Training

This can be achieved through points like knowledge distillation techniques and Parameter-Efficient Fine-Tuning (PEFT). Knowledge distillation techniques, using methods like Soft Labels and intermediate layer feature transfer (e.g., FitNets), allow large models to transfer implicit knowledge (such as semantic representation, reasoning logic) into training signals for small models. This enables small models to inherit the core capabilities of large models while remaining lightweight (e.g., TinyBERT achieves 96% of BERT's performance on the GLUE benchmark with 75% fewer parameters).

Parameter-Efficient Fine-Tuning, through techniques like LoRA and QLoRA, allows small models to perform localized fine-tuning based on the pre-trained parameters of large models, quickly adapting to specific tasks (e.g., in medical Q&A scenarios, the accuracy of small models increased by 20% after fine-tuning). This forms a collaborative training paradigm of "large model foundation + small model specialization."

2.3.4 Adaptability of Deployment Scenarios

In terms of adaptability, collaboration between cloud and edge can be utilized. Large models rely on high-compute cloud servers (requiring tens of GB of VRAM) and are suitable for handling batch complex tasks (e.g., text generation, multi-turn dialogue). Small models can be deployed on low-compute devices (e.g., Raspberry Pi, mobile phone chips) and are responsible for real-time interaction and local data preprocessing (e.g., speech recognition, preliminary user intent analysis). The two communicate via APIs or lightweight protocols (e.g., gRPC) to enable data flow.

2.3.3 架构与训练的可迁移性

这可通过知识蒸馏技术、参数高效微调等要点实现。知识蒸馏技术通过软标签、中间层特征迁移（如 FitNets）等方法，使大模型能够将隐式知识（如语义表示、推理逻辑）转化为小模型的训练信号，从而让小模型在保持轻量化的同时继承大模型的核心能力（例如，TinyBERT 在 GLUE 基准上以参数量减少 75% 达到 BERT 性能的 96%）。

参数高效微调通过 LoRA、QLoRA 等技术，让小模型能够基于大模型的预训练参数进行局部微调，快速适应特定任务（例如，在医疗问答场景中，小模型经微调后准确率提升 20%）。这形成了“大模型基础 + 小模型专精”的协同训练范式。

2.3.4 部署场景的适应性

在适应性方面，可利用云边协同。大模型依赖高算力云服务器（需数十 GB 显存），适合处理批量复杂任务（如文本生成、多轮对话）。小模型可部署于低算力设备（如树莓派、手机芯片），负责实时交互与本地数据预处理（如语音识别、初步用户意图分析）。二者通过 API 或轻量协议（如 gRPC）进行通信，实现数据流转。

2.3.5 协作的必要性：现实需求驱动的必然选择

在大模型与小模型的协作过程中，应对现实需求至关重要。需要考虑的问题包括计算成本约束、实时性要求、某些层级任务的处理以及多模态性等。

具体而言，在计算能力方面，大模型单次推理的能耗是小模型的数百倍（例如，GPT-4 生成 1000 字消耗的能量约为 TinyBERT 的 500 倍）。在高频交互场景中（例如客服、实时翻译），仅依赖大模型将导致成本飙升（企业级应用月度计算成本可能超过百万元）。此外，边缘设备（例如自动驾驶系统）要求响应延迟低于 50 毫秒。基于云端的大模型推理延迟通常超过 500 毫秒，无法满足需求。小模型的本地部署可实现亚毫秒级响应。与大模型协作可将端到端延迟压缩至 200 毫秒以内（例如智能汽车中的语音交互系统）。复杂任务（例如法律文件生成、科学论文润色）由大模型处理，而简单任务（例如关键词提取、情感分类）则由小模型处理，形成“金字塔式”任务分配体系。在电商客服中，小模型首先识别用户意图（准确率 90

小模型可对图像、语音等多模态数据进行预处理（例如语音转文本、图像特征提取）。大模型执行跨模态推理（例如根据用户语音和表情生成个性化回应），从而降低大模型的输入复杂度（例如多模态处理延迟降低 30）。医疗、金融等领域要求用户数据保留在本地。小模型可在设备上数据进行清洗和匿名化处理（例如从医疗记录中移除姓名和身份证号），仅将匿名化特征传输给大模型，从而避免隐私泄露风险（例如在联邦学习中，小模型在本地训练参数，而大模型聚合全局知识）。小模型结构简单（例如单层 Transformer 或 CNN），其决策逻辑可通过可视化工具进行解释（例如注意力热力图），弥补了大模型的“黑箱”特性（例如在医疗诊断场景中，小模型提供初步结论，大模型进行验证并生成解释性文本）。

除了上述问题，应用场景也需要考虑。大模型难以适应小众领域（例如方言识别、垂直行业知识库）。小模型可针对特定场景进行定制（例如税务领域的小模型仅需 10 亿参数，在政策解读中达到 95SLMs（小型语言模型）凭借其高效性和特异性，已成为人工智能系统中不可

2.3.5 Necessity of Collaboration: An Inevitable Choice Driven by Real-world Needs

In the collaboration process between large and small models, addressing real-world needs is essential. Issues to consider include computational cost constraints, real-time requirements, handling of certain hierarchical tasks, and multimodality, among others.

Specifically, regarding computational power, a single inference by a large model consumes hundreds of times more power than a small model (e.g., GPT-4 generating 1000 words consumes about 500 times the energy of TinyBERT). In high-frequency interaction scenarios (e.g., customer service, real-time translation), relying solely on large models would lead to soaring costs (enterprise-level application monthly compute costs could exceed millions of yuan). Furthermore, edge devices (e.g., autonomous driving systems) require response latency below 50ms. Cloud-based inference latency for large models is typically over 500ms, failing to meet the demand. Local deployment of small models can achieve sub-millisecond responses. Collaboration with large models can compress end-to-end latency to within 200ms (e.g., voice interaction systems in smart cars). Complex tasks (e.g., legal document generation, scientific paper polishing) are handled by large models, while simple tasks (e.g., keyword extraction, sentiment classification) are processed by small models, forming a "pyramid-style" task allocation system. In e-commerce customer service, a small model first identifies user intent (90% accuracy); if it's a complex query (e.g., interpreting return policies), it forwards it to the large model, increasing efficiency by 40% compared to a single-model approach.

Small models can preprocess data from modalities like images and speech (e.g., speech-to-text, image feature extraction). Large models perform cross-modal reasoning (e.g., generating personalized responses based on user voice and expression), reducing the input complexity for large models (e.g., multimodal processing latency reduced by 30%). Fields like healthcare and finance require user data to remain local. Small models can perform data cleaning and anonymization on the device (e.g., removing names and IDs from medical records), transmitting only anonymized features to the large model, thus avoiding privacy leakage risks (e.g., in federated learning, small models train parameters locally, while the large model aggregates global knowledge). The simple structure of small models (e.g., single-layer Transformer or CNN) allows their decision logic to be explained using visualization tools (e.g., attention heatmaps), compensating for the "black box" nature of large models (e.g., in medical diagnosis scenarios, the small model provides a preliminary conclusion, which the large model verifies and generates an explanatory text).

Beyond these issues, application scenarios need consideration. Large models struggle to adapt to niche domains (e.g., dialect recognition, vertical industry knowledge bases). Small models can be customized for specific scenarios (e.g., a small model for the tax domain with only 1 billion parameters achieves 95% accuracy in policy interpretation), forming a collaborative ecosystem of "general foundation + domain plugin" with large models. Small and medium-sized enterprises and developers often cannot afford the computational power and training costs of large models. By combining small models with cloud-based large model APIs (e.g., using OpenAI's GPT-4 API with a local small model for pre-filtering), deployment costs can be reduced by over 90%, promoting the democratization of AI technology.

SLMs often handle the front-end tasks in a model pipeline, such as input understanding, candidate generation, or decision-making, making them a crucial part of achieving efficient collaboration (gao2025strategic). In complex systems, SLMs handle front-end basic tasks (like input understanding, candidate generation), passing the preliminary processed results to large models (LLMs) for deep reasoning, forming a collaboration mode of "lightweight model preprocessing + heavyweight model refinement." In intelligent customer service systems: An SLM quickly analyzes the user's question ("check order status"), extracts key information

或缺的“前端处理器”。通过与 LLMs 的分工协作，它们显著提升了整体性能和资源利用率。这种架构在工业界得到广泛应用，是实现“大模型 + 小模型”混合生态系统的重要实践。

3 大模型与小模型的协作机制与架构

3.1 协作模式分类

根据大语言模型 (LLM) 与小语言模型 (SLM) 之间的交互方式和信息流向，其协作模式可分为流水线式、混合/路由式、辅助/增强式、知识蒸馏驱动式以及集成/融合式协作。

3.1.1 流水线协作

流水线协作是一种顺序执行模式，其中一个模型的输出作为另一个模型的输入 (wang2024comprehensive)。由于 SLM 通常具有更高的效率，它们常被用于执行初步处理或生成候选结果，然后传递给 LLM 进行更复杂的推理或知识整合 (gao2025strategic)。

在推荐系统中，流水线协作被广泛使用。LLM 可以根据用户历史行为和偏好生成一系列候选推荐项目，这利用了 LLM 捕捉用户偏好的强大能力 (lv2025collaboration)。然后，部署在本地设备上的 SLM 可以根据用户的实时交互行为对这些候选项目进行重新排序，从而更准确地反映用户当前的兴趣。这种方法利用了 LLM 的全局理解能力，同时也兼顾了 SLM 对实时数据的快速响应能力 (lv2025collaboration)。

CoGenesis 框架是流水线协作的另一个典型示例，如图 1 所示。在该框架中，部署在用户本地设备上的 SLM 可以访问用户的私有数据和活动日志，并基于此信息处理指令。对于需要更深层次推理的任务，SLM 的输出可以作为输入传递给部署在云基础设施中的 LLM。这种设计保护了用户隐私，同时利用了大型模型的强大能力。此外，SLM 可用于从输入文本中提取关键信息或生成简洁的提示，然后发送给 LLM，以指导其生成更相关的输出 (zhang2024cogenesis; wang2024comprehensive)。

流水线协作的关键在于合理规划任务并设计有效的模型间接口，以确保信息能够在不同模型之间准确、高效地传递 (wang2024comprehensive; liu2025towards; chen2025knowledge; xu2024slmrec)。这种模式的有效性在很大程度上取决于第一个模型提取相关信息的能力以及传递给第二个模型的信息质量。如果 SLM 未能准确捕捉关键信息，或者以不适当的格式传递信息，LLM 的性能将受到不利影响。因此，必须仔细考虑通信协议和交换信息的类型，以确保无缝且有效的协作。流水线协作通常利用 SLM 的效率进行初步处理或上下文收集，将计算密集型 and 知识依赖型任务卸载给 LLM，从而在速度与准确性之间取得平衡。

3.1.2 混合/路由协作

混合或路由协作指的是使用一种机制（通常称为路由器）来决定哪个模型 (LLM 或 SLM) 应该处理特定的输入或子任务 (wang2025mixllm; zheng2025citer; lv2025collaboration)。路由决策通常基于任务复杂度、领域、成本、所需延迟或其他预定义的标准。

CITER 框架是路由协作的一个典型示例，如图 2 所示。该框架采用令牌级路由策略，将非关键令牌路由到 SLM 以提高效率，同时将关键令牌路由到 LLM 以确保生成质量。这种细

(order number), and generates candidate query solutions. The LLM then uses these candidate solutions to call database interfaces and generate the final response. SLMs (Small Language Models), with their efficiency and specificity, have become indispensable "front-end processors" in AI systems. Through division of labor with LLMs, they significantly improve overall performance and resource utilization. This architecture is widely applied in industry and is an important practice for realizing the "Large Model + Small Model" hybrid ecosystem.

3 Collaboration Mechanisms and Architectures for Large and Small Models

3.1 Classification of Collaboration Modes

Based on the interaction methods and information flow between LLMs and SLMs, their collaboration modes can be classified into pipeline, hybrid/routing, auxiliary/enhancement, knowledge distillation-driven, and integration/fusion collaboration.

3.1.1 Pipeline Collaboration

Pipeline collaboration is a sequential execution mode where the output of one model serves as the input for another (wang2024comprehensive). Since SLMs typically offer higher efficiency, they are often used for executing preliminary processing or generating candidate results, which are then passed to LLMs for more complex reasoning or knowledge integration (gao2025strategic).

In recommendation systems, pipeline collaboration is widely used. An LLM can generate a series of candidate recommendation items based on user historical behavior and preferences, leveraging the LLM's strong ability to capture user preferences (lv2025collaboration). Then, an SLM deployed on the local device can re-rank these candidate items based on the user's real-time interaction behavior, thereby more accurately reflecting the user's current interests. This approach utilizes the LLM's global understanding capabilities while also accommodating the SLM's rapid response to real-time data (lv2025collaboration).

The CoGenesis framework is another typical example of pipeline collaboration, as shown in Figure 1. In this framework, an SLM deployed on the user's local device can access the user's private data and activity logs and process instructions based on this information. For tasks requiring deeper reasoning, the SLM's output can be passed as input to an LLM deployed in cloud infrastructure. This design protects user privacy while leveraging the powerful capabilities of large models. Additionally, SLMs can be used to extract key information from input text or generate concise prompts, which are then sent to LLMs to guide them in generating more relevant outputs (zhang2024cogenesis; wang2024comprehensive).

The key to pipeline collaboration lies in reasonably dividing tasks and designing effective inter-model interfaces to ensure that information can be accurately and efficiently transferred between different models (wang2024comprehensive; liu2025towards; chen2025knowledge; xu2024slmrec). The effectiveness of this mode largely depends on the ability of the first model to extract relevant information and the quality of the information passed to the second model. If the SLM fails to accurately capture key information or passes information in an inappropriate format, the LLM's performance will be adversely affected. Therefore, communication protocols and the type of information exchanged must be carefully considered to ensure seamless and effective collaboration. Pipeline collaboration typically leverages the efficiency of SLMs for preliminary processing or context collection, offloading computationally intensive and knowledge-dependent tasks to LLMs, thereby striking a balance between speed and accuracy.

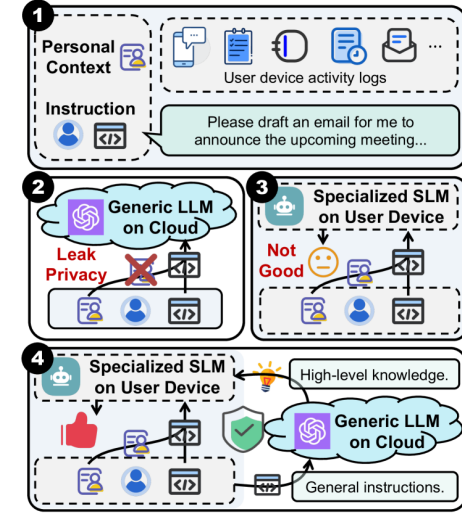


Figure 1: CoGenesis 框架结构图。1. 上下文感知的指令示例。2. 上下文感知的大语言模型（LLM）在上下文感知方面表现出色，但存在隐私风险。3. 设备端专用小语言模型（SLM）优先考虑隐私，但性能较低。4. 协作的 LLM 和 SLM 增强了隐私保护并提高了性能。

粒度的路由允许系统在效率和质量之间动态调整 (zheng2025citer)。

其他研究工作侧重于开发能够从候选模型池中选择最合适 LLM 的路由器 (varangot2025doing; wang2025mixllm)。这种方法可以扩展到包含 SLM，使路由器能够根据任务需求在不同规模和能力的模型中进行选择。HybridLLM 利用二元分类器来预测查询难度，并相应地路由到不同模型 (yao2025toward)。

级联是一种常见的路由策略，首先尝试使用较小的 LLM 处理输入，并在必要时将查询传递给较大的 LLM (chen2025harnessing; ong2024routellm)。这种方法可以在确保性能的同时降低计算成本。

智能路由需要一种能够准确评估输入特征和可用模型能力以做出最优决策的机制。路由器需要理解任务的复杂性、查询的领域以及系统中每个 LLM 和 SLM 的具体优势和劣势。这可能涉及训练一个单独的模型来预测性能，或使用基于查询特征的启发式方法。在设计路由机制时，平衡成本、延迟和质量是一个关键的考虑因素。更强大的模型通常伴随着更高的成本和延迟，因此路由策略必须根据应用需求权衡这些因素。对于实时应用，延迟可能是最关键的因素，可能会优先为某些查询类型选择更快（但可能准确性较低）的 SLM。对于需要高准确性的任务，系统可能会路由到更强大（也更昂贵）的 LLM。路由策略需要可配置，以满足不同的性能目标。

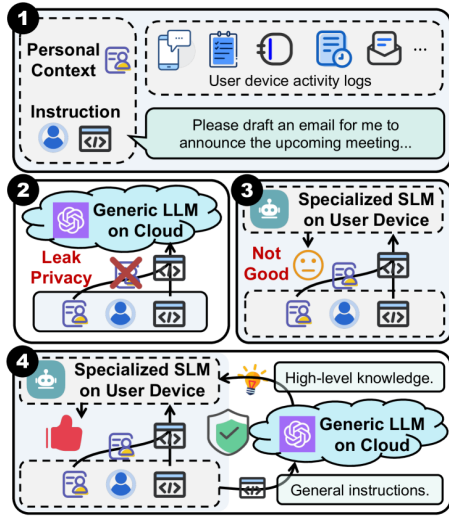


Figure 1: CoGenesis framework structure diagram. 1. Context-aware instructional examples. 2. Context-aware Language Models (LLMs) excel in context awareness but pose privacy risks. 3. On-device specialized Small Language Models (SLMs) prioritize privacy but have lower performance. 4. Collaborative LLMs and SLMs enhance privacy protection and improve performance.

3.1.2 Hybrid/Router Collaboration

Hybrid or routing collaboration refers to using a mechanism (usually called a router) to decide which model (LLM or SLM) should handle a specific input or sub-task ([wang2025mixllm](#); [zheng2025citer](#); [lv2025collaboration](#)). The routing decision is typically based on task complexity, domain, cost, required latency, or other predefined criteria.

The CITER framework is a typical example of routing collaboration, shown in Figure 2. This framework adopts a token-level routing strategy, routing non-critical tokens to the SLM to improve efficiency, while routing critical tokens to the LLM to ensure generation quality. This fine-grained routing allows the system to dynamically adjust between efficiency and quality ([zheng2025citer](#)).

Other research efforts focus on developing routers capable of selecting the most appropriate LLM from a pool of candidate models ([varangot2025doing](#); [wang2025mixllm](#)). This approach can be extended to include SLMs, enabling the router to choose among models of different scales and capabilities based on task requirements. HybridLLM utilizes a binary classifier to predict query difficulty and routes between different models accordingly ([yao2025toward](#)).

Cascading is a common routing strategy that first attempts to process input using smaller LLMs and passes the query to larger LLMs if necessary ([chen2025harnessing](#); [ong2024routellm](#)). This method can reduce computational costs while ensuring performance.

Intelligent routing requires a mechanism capable of accurately assessing input features

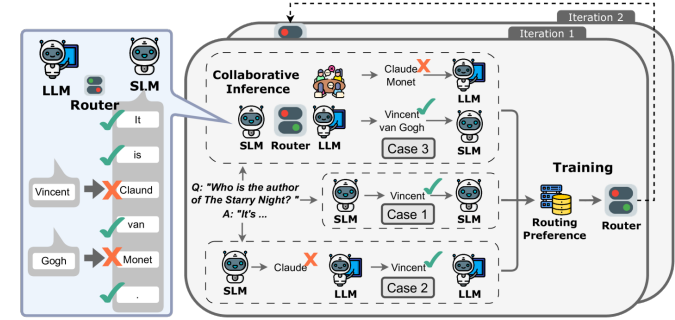


Figure 2: CITER 框架结构图。利用路由器在 SLM 和 LLM 之间进行协作推理。路由器使用通过三种场景收集的路由偏好进行训练。场景 1：SLM 生成正确的令牌，路由偏好分配给 SLM。场景 2：SLM 生成错误的令牌，而 LLM 生成正确的令牌，路由偏好分配给 LLM。场景 3：SLM 和 LLM 都未生成正确的令牌；执行协作推理以获得完整响应以分配路由偏好。

3.1.3 辅助/增强协作

在辅助或增强协作中，一个模型（可以是 LLM 或 SLM）辅助另一个模型，以提升其性能或能力 ([wang2024comprehensive](#); [shao2025division](#); [hu2024auxiliary](#))。

LLM 可以将一个复杂查询分解为若干子问题，然后将这些子问题分配给 SLM 处理，反之亦然 ([shao2025division](#); [xu2025collab](#))。SLM 可以为 LLM 提供上下文信息或领域特定知识，以增强 LLM 的推理能力。LLM 也可用于为 SLM 生成训练数据或提供反馈信号，从而帮助 SLM 学习与改进 ([deng2023mutual](#))。

Collab-RAG 框架展示了辅助协作的一个应用，如图 3 所示。在此框架中，一个 SLM 负责将用户查询分解为更简单的子问题，以便于从知识库中检索相关信息，从而增强 LLM 在回答复杂问题时的推理能力 ([xu2025collab](#))。

辅助协作允许基于每种模型类型固有优势进行分工，与单独使用任一模型相比，可能提升整体性能。LLM 擅长理解复杂指令并拥有广泛知识，而 SLM 在某些特定任务上可能更专业、更高效。通过让它们相互辅助，系统可以利用 LLM 的理解能力来指导 SLM，或利用 SLM 的专业知识来为 LLM 的推理提供信息。设计有效的辅助机制需要仔细考虑模型如何能最好地互补，以及用于这种交互的通信方法。简单地让一个模型将其输出馈送给另一个模型可能不是最有效的方法。辅助可能涉及提供特定类型的信息、指导推理过程，或对另一个模型的性能提供反馈。通信需要根据所提供的具体辅助进行定制。

3.1.4 知识蒸馏驱动的协作

知识蒸馏 (KD) 是一种关键技术，用于将知识和能力从大型（通常是专有的）教师模型 (LLM) 转移到更小、更高效的学生模型 (SLM) ([xu2024survey](#); [xu2024slmrec](#))。

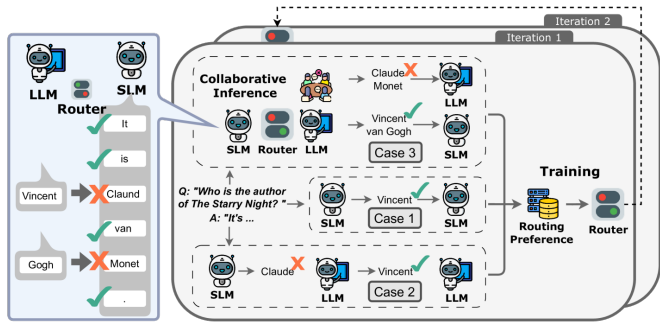


Figure 2: CITER framework structure diagram. Utilizes a router for collaborative inference between SLM and LLM. The router is trained using routing preferences collected through three scenarios. Scenario 1: SLM generates the correct token, routing preference assigned to SLM. Scenario 2: SLM generates an incorrect token, while LLM generates the correct token, routing preference assigned to LLM. Scenario 3: Neither SLM nor LLM generates the correct token; collaborative reasoning is performed to obtain the complete response for assigning routing preference.

and the capabilities of available models to make optimal decisions. The router needs to understand the complexity of the task, the domain of the query, and the specific strengths and weaknesses of each LLM and SLM in the system. This might involve training a separate model to predict performance or using heuristic methods based on query features. Balancing cost, latency, and quality is a key consideration when designing routing mechanisms. More powerful models typically come with higher costs and latency, so the routing strategy must weigh these factors according to the application's needs. For real-time applications, latency might be the most critical factor, potentially prioritizing faster (but possibly less accurate) SLMs for certain query types. For tasks requiring high accuracy, the system might route to a more powerful (and more expensive) LLM. Routing strategies need to be configurable to meet different performance objectives.

3.1.3 Auxiliary/Enhancement Collaboration

In auxiliary or enhancement collaboration, one model (either LLM or SLM) assists another to improve its performance or capabilities (wang2024comprehensive; shao2025division; hu2024auxiliary).

An LLM can decompose a complex query into several sub-problems and then assign these sub-problems to an SLM for processing, or vice versa (shao2025division; xu2025collab). SLMs can provide LLMs with contextual information or domain-specific knowledge to enhance the LLMs' reasoning capabilities. LLMs can also be used to generate training data or provide feedback signals for SLMs, thereby helping SLMs learn and improve (deng2023mutual).

The Collab-RAG framework demonstrates an application of auxiliary collaboration, as shown in Figure 3. In this framework, an SLM is responsible for decomposing user queries into simpler sub-problems to facilitate the retrieval of relevant information from a knowledge base, thereby enhancing the LLM's reasoning ability when answering complex questions (xu2025collab).

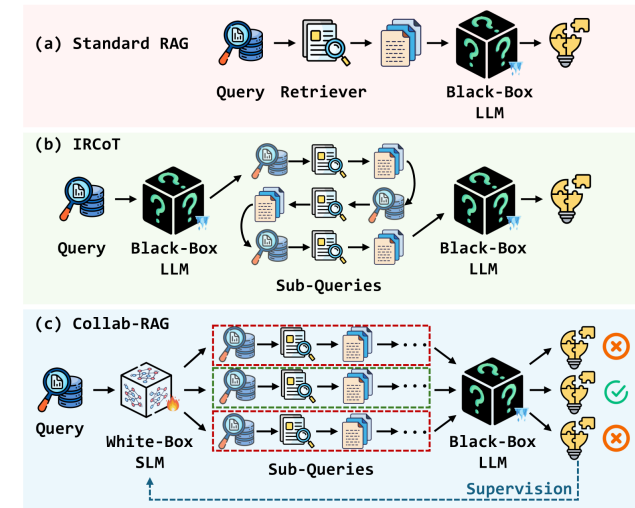


Figure 3: Collab-RAG 的迭代训练框架。SLM 根据 LLM 阅读器的生成质量更新其参数。此过程迭代多次，逐步提升 SLM 的分解能力。

KD 涉及多个方面，包括知识类型（输出概率、中间表示）、蒸馏算法（监督微调、散度最小化）以及应用领域（模型压缩、技能迁移、领域专业化）。数据增强在提升 LLM 知识蒸馏效果方面也扮演着重要角色。根据教师模型能否提供内部信息，KD 方法可分为白盒 KD 和黑盒 KD。KD 也用于迁移特定技能，例如推理、指令遵循和工具使用 (xu2024survey; hendriks2025honey; gu2023minillm)。

知识蒸馏是通过创建更小、更高效的模型，使 LLM 级别的智能能够在资源受限环境中部署的关键机制。虽然 LLM 提供了卓越的性能，但其规模和计算需求限制了许多实际场景中的应用。KD 允许我们将这些大型模型学到的知识转移到更小的模型中，这些模型可以部署在边缘设备上，或用于具有严格延迟要求的应用。知识蒸馏的成功取决于几个因素，包括教师和学生模型的选择、蒸馏数据的质量和数量，以及所采用的具体蒸馏技术。不同的 LLM 具有不同的优势，教师模型的选择将影响所迁移知识的类型。蒸馏数据应能代表学生模型将要执行的任务。选择合适的蒸馏算法并调整其参数对于实现最佳性能也至关重要。

3.1.5 集成/融合协作

集成或融合协作是指将 LLM 和 SLM 的架构或输出结合成一个统一的系统，以更紧密耦合的方式利用它们的互补优势 (naveed2023comprehensive; subramanian2025small; lv2025collaboration; dong2024hymba; wan2024knowledge)。

Hymba 是一种混合架构，如图 4所示，它在同一层内集成了 Transformer 的注意力机制

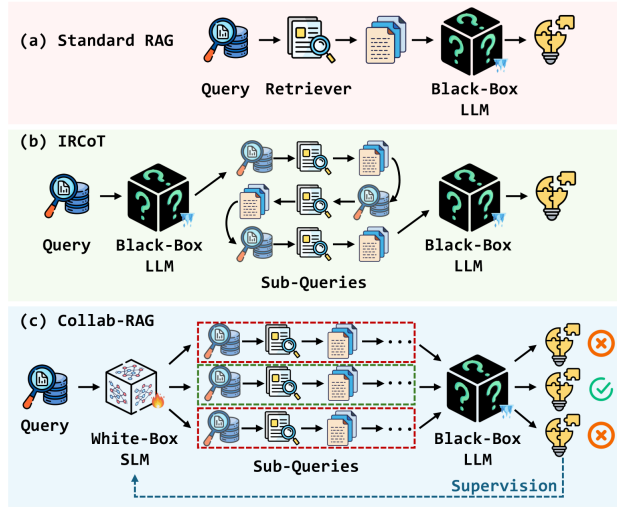


Figure 3: Iterative training framework of Collab-RAG. The SLM updates its parameters based on the generation quality of the LLM reader. This process iterates multiple times, progressively improving the SLM’s decomposition capability.

Auxiliary collaboration allows for a division of labor based on the inherent strengths of each model type, potentially improving overall performance compared to using either model alone. LLMs excel at understanding complex instructions and possess broad knowledge, while SLMs may be more specialized and efficient on certain specific tasks. By having them assist each other, the system can leverage the LLM’s understanding to guide the SLM, or use the SLM’s specialized knowledge to inform the LLM’s reasoning. Designing effective auxiliary mechanisms requires careful consideration of how the models can best complement each other and the communication methods used for this interaction. Simply having one model feed its output to another might not be the most effective approach. Assistance might involve providing specific types of information, guiding the reasoning process, or offering feedback on the other model’s performance. Communication needs to be tailored to the specific assistance being provided.

3.1.4 Knowledge Distillation-Driven Collaboration

Knowledge Distillation (KD) is a key technique used to transfer knowledge and capabilities from a large, often proprietary teacher model (LLM) to a smaller, more efficient student model (SLM) (xu2024survey; xu2024slmrec).

KD involves multiple aspects, including the type of knowledge (output probabilities, intermediate representations), distillation algorithms (supervised fine-tuning, divergence minimization), and application domains (model compression, skill transfer, domain specialization). Data augmentation also plays an important role in enhancing the effectiveness of KD for LLMs. Depending on whether the teacher model can provide internal information, KD methods can

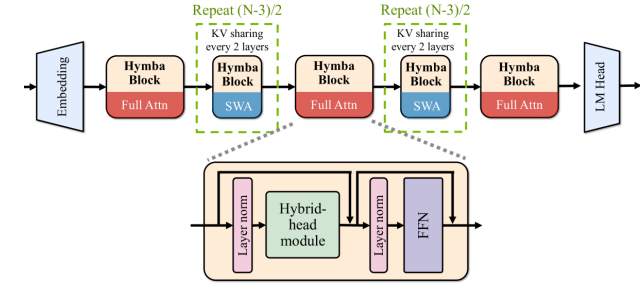


Figure 4: Hymba 模型的整体架构

和状态空间模型 (SSM)，从而实现了高召回率和高效的上下文摘要 (dong2024hymba)。模型融合技术是指结合多个模型的参数或预测结果。这可以包括权重平均、集成方法或更复杂的融合架构。多模态大语言模型 (MLLM) 是集成协作的另一个例子，通常使用 LLM 作为主干，并整合来自不同模态（如文本和图像）的信息。

集成协作通过在架构层面深度结合不同类型模型的优势，为创建更强大、更多功能的 AI 系统提供了潜力。模型融合提供了一种经济高效的方法，通过结合现有预训练模型的知识 and 能力来创建更强大的模型，而无需进行大量的重新训练。然而，有效融合具有不同架构的模型，并确保所得模型保留所需的特性，仍然具有挑战性。

3.2 实现协作的关键技术

为了有效实现 LLM 和 SLM 之间的协作，需要以下关键技术：

3.2.1 任务分配与智能路由

在异构的 LLM-SLM 系统中，任务分配面临诸多挑战，需要考虑任务复杂性、所需知识、计算资源、延迟约束和成本 (varangot2025doing)。主要目标是智能地将系统输入任务分配给最合适的模型进行处理。

通过动态的复杂度感知路由技术，系统可以根据任务特征（如任务类型、上下文长度、领域特异性）和输入复杂性来决定使用哪个模型。在 Hybrid LLM 等系统中，路由器根据查询难度和期望的质量水平做出决策 (ding2024hybrid)。

自我强化的路由优化利用强化学习逐步改进路由策略，以模型的实际性能作为反馈信号，持续优化和调整路由策略。MixLLM 使用上下文感知的上下文老虎机算法来学习最优的查询-模型分配，从而适应不断变化的需求和模型能力 (wang2025mixllm)。

智能路由机制可以根据预定义的标准或学习到的策略，动态地将任务分配给最合适的模型。这包括基于查询难度、性能预测和成本意识的路由 (wang2025mixllm; mohammadshahi2024routoo)。更小、更快的模型可用于初始意图检测，将请求路由到更专业化或更大的模型 (guha2024smoothie)。有效的任务分配和路由对于优化协作 LLM-SLM 系统的性能和效率至关重要，确保每项任务

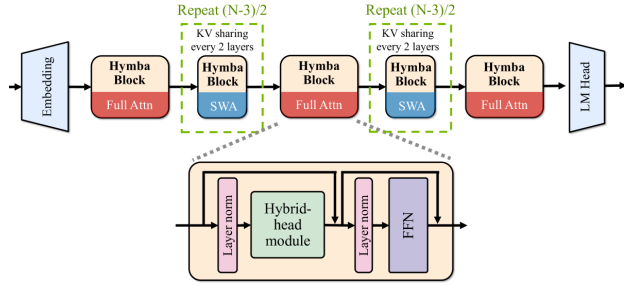


Figure 4: Overall architecture of the Hymba model

be divided into white-box KD and black-box KD. KD is also used to transfer specific skills, such as reasoning, instruction following, and tool use (xu2024survey; hendriks2025honey; gu2023minillm).

Knowledge distillation is a critical mechanism for enabling LLM-level intelligence to be deployed in resource-constrained environments by creating smaller, more efficient models. While LLMs offer superior performance, their size and computational requirements limit their applicability in many practical scenarios. KD allows us to transfer the knowledge learned by these large models into smaller models that can be deployed on edge devices or used in applications with strict latency requirements. The success of knowledge distillation depends on several factors, including the choice of teacher and student models, the quality and quantity of distillation data, and the specific distillation techniques employed. Different LLMs have different strengths, and the choice of teacher model will influence the type of knowledge transferred. The distillation data should be representative of the tasks the student model will perform. Selecting the appropriate distillation algorithm and tuning its parameters are also crucial for achieving optimal performance.

3.1.5 Integration/Fusion Collaboration

Integration or fusion collaboration refers to combining the architectures or outputs of LLMs and SLMs into a unified system to leverage their complementary strengths in a more tightly coupled manner (naveed2023comprehensive; subramanian2025small; lv2025collaboration; dong2024hymba; wan2024knowledge).

Hymba is a hybrid architecture, shown in Figure 4, which integrates the attention mechanism of transformers and state-space models (SSMs) within the same layer, thereby achieving high recall and efficient context summarization (dong2024hymba). Model fusion techniques refer to combining the parameters or predictions of multiple models. This can include weight averaging, ensemble methods, or more complex fusion architectures. Multimodal LLMs (MLLMs) are another example of integration collaboration, typically using an LLM as a backbone and integrating information from different modalities (such as text and images).

Integration collaboration offers the potential to create more powerful and versatile AI systems by deeply combining the strengths of different model types at the architectural level. Model fusion provides a cost-effective way to create more powerful models without extensive retraining by combining the knowledge and capabilities of existing pre-trained models. However, effectively fusing models with different architectures and ensuring the resulting model retains desired properties remain challenging.

都由最合适的模型处理。开发稳健且自适应的路由机制需要深入了解每个模型的能力以及所处理任务的特征。这可能涉及创建基准来评估模型在不同任务类型上的性能，并设计能够适应变化条件的路由策略。

3.2.2 模型间通信与接口设计

在协作环境中，大语言模型（LLM）与小语言模型（SLM）之间无缝且高效的通信至关重要。

自然语言接口在此过程中扮演着重要角色，特别是在智能体链（Chain-of-Agents）(zhang2024chain) 中，工作智能体可以通过自然语言传递信息。这种方法利用了 LLM 的语言理解能力，简化了接口设计的复杂性，并提高了系统的可解释性 (zhang2024chain)。

结构化的中间表示可以提供更标准化的通信机制。CoGenesis (zhang2024cogenesis) 使用“草图”（sketch）格式在大模型与小模型之间进行通信。这种表示以结构化摘要的形式传达关键信息，使得下游模型更易于处理，并且比原始文本更紧凑且信息丰富 (zhang2024cogenesis)。

通过概率分布共享，模型可以在更深层次上传递不确定性。在 FS-GEN (zhang2024fast) 和 CoGenesis (zhang2024fast) 中，logits 共享允许大模型将词元概率分布传递给小模型。这不仅传递了最终预测，还传递了选择的相对可能性，使得小模型能够通过将此信息与本地上下文结合起来做出更明智的决策 (zhang2024cogenesis; zhang2024fast)。

这涉及不同的通信方法，例如传递基于文本的提示和响应、共享中间表示（如嵌入、隐藏状态），或使用专门的 API 进行模型间通信 (king2024thoughtful; pozdniakov2024large; wozniak2024personalized)。设计标准化的接口，使得不同模型无论其底层架构如何都能有效交互，也是一个重要的考量。此外，还需要解决确保模型间数据一致性和格式兼容性相关的挑战。协作的效率和效果直接受到模型间通信和信息交换程度的影响。精心设计的接口和通信协议对于实现无缝交互至关重要。通信方法的选择和接口设计应根据具体的协作模式和信息交换类型进行定制。流水线协作可能只需要传递最终输出，而更紧密集成的模型则可能受益于共享中间表示。

3.2.3 模型融合与结果整合

模型融合是指合并 LLM 和 SLM 的架构或参数以创建更优的模型，包括权重平均、知识融合、集成学习、基于概率分布的融合、模型堆叠以及专家混合模型等技术 (tang2024small; wan2024knowledge; yang2024model; mavromatis2024pack; wang2025speculate; shi2024profuser)。

结果整合是指合并协作模型的输出。常见方法包括简单平均、加权平均、多数投票，或使用另一个模型来组合输出。此外，还需要处理不同模型输出之间潜在的不一致或冲突。模型融合与结果整合对于创建能够利用多个模型独特优势和知识的统一系统至关重要。技术的选择取决于协作的具体目标以及所涉及模型的特点。评估模型融合与结果整合的有效性需要仔细考虑适当的指标和基准，以评估组合系统的性能。

3.2 Key Technologies for Implementing Collaboration

To effectively implement collaboration between LLMs and SLMs, the following key technologies are required:

3.2.1 Task Allocation and Intelligent Routing

In heterogeneous LLM-SLM systems, task allocation faces numerous challenges, requiring consideration of task complexity, required knowledge, computational resources, latency constraints, and cost (**varangot2025doing**). The main goal is to intelligently assign system input tasks to the most appropriate model for processing.

Through dynamic complexity-aware routing techniques, the system can decide which model to use based on task characteristics (such as task type, context length, domain specificity) and input complexity. In systems like Hybrid LLM, the router makes decisions based on query difficulty and desired quality level (**ding2024hybrid**).

Self-reinforcing routing optimization utilizes reinforcement learning to progressively improve routing strategies, using the actual performance of the models as feedback signals to continuously optimize and adjust routing policies. MixLLM uses a context-aware contextual-bandit algorithm to learn the optimal query-model allocation, thereby adapting to changing demands and model capabilities (**wang2025mixllm**).

Intelligent routing mechanisms can dynamically assign tasks to the most suitable model based on predefined criteria or learned policies. This includes routing based on query difficulty, performance prediction, and cost awareness (**wang2025mixllm**; **mohammadshahi2024routoo**). Smaller, faster models can be used for initial intent detection, routing requests to more specialized or larger models (**guha2024smoothie**). Effective task allocation and routing are crucial for optimizing the performance and efficiency of collaborative LLM-SLM systems, ensuring each task is handled by the most appropriate model. Developing robust and adaptive routing mechanisms requires a deep understanding of each model's capabilities and the characteristics of the tasks being processed. This might involve creating benchmarks to evaluate model performance on different task types and designing routing strategies that can adapt to changing conditions.

3.2.2 Inter-Model Communication and Interface Design

Seamless and efficient communication between LLMs and SLMs is crucial in collaborative environments.

Natural language interfaces play an important role in this process, particularly in Chain-of-Agents (**zhang2024chain**), where working agents can pass information via natural language. This approach leverages the language understanding capabilities of LLMs, simplifies the complexity of interface design, and improves system interpretability (**zhang2024chain**).

Structured intermediate representations can provide a more standardized communication mechanism. CoGenesis (**zhang2024cogenesis**) uses a 'sketch' format for communication between large and small models. This representation conveys key information as a structured summary, making it easier for downstream models to process and more compact yet information-rich than raw text (**zhang2024cogenesis**).

Through probability distribution sharing, models can communicate uncertainty at a deeper level. In FS-GEN (**zhang2024fast**) and CoGenesis (**zhang2024fast**), logits sharing allows large models to pass token probability distributions to small models. This not only conveys the final prediction but also the relative likelihood of choices, allowing the small model to make more informed decisions by combining this information with local context (**zhang2024cogenesis**; **zhang2024fast**).

3.2.4 状态同步与上下文管理

在协作式大语言模型与小语言模型中，尤其是在多轮对话或序列任务中，保持状态一致性和管理上下文至关重要。这涉及同步协作模型间的内部状态（记忆、注意力权重）以及管理和共享它们之间的上下文信息 (**subramanian2025small**; **wang2024comprehensive**; **naveed2023comprehensive**; **mojarradi2024improving**)，例如使用共享记忆模块或在输入时传递上下文。此外，还需要解决与处理长上下文场景以及确保跨多次交互的上下文连贯性相关的挑战。

保持状态一致性和有效管理上下文对于确保协作式大语言模型-小语言模型系统进行连贯且有意义的交互至关重要，特别是对于涉及多个步骤或轮次的任务。状态同步和上下文管理的复杂性随着协作模型数量和交互长度的增加而增加。开发可扩展且高效的技术来处理这些方面是一个持续的研究挑战。

3.2.5 动态资源调度与优化

在协作式大语言模型-小语言模型系统中，尤其是在处理动态工作负载和变化的资源需求时，高效分配和管理计算资源（如 GPU 内存、处理时间）是一个重大挑战 (**wang2024comprehensive**)。

动态资源调度技术可以根据协作模型的当前需求和可用资源来调整资源分配。优化策略，如量化、剪枝和高效注意力机制，可用于降低协作推理的计算成本和内存占用 (**huang2025dynamic**; **amayuelas2025self**; **sun2024illumnix**)。云计算与边缘计算的结合也为资源管理提供了潜力，例如，大语言模型可以在云端运行，而小语言模型在边缘设备上运行 (**lv2025collaboration**)。高效的资源调度和优化对于使协作式大语言模型-小语言模型系统变得实用且具有成本效益至关重要，尤其是在资源受限或高流量的现实应用中。大语言模型和小语言模型在资源需求和性能特征方面的异构性增加了资源调度和优化的复杂性。策略需要考虑这些差异以实现最优的资源利用。

4 由设备端需求驱动的应用场景

随着大语言模型的快速发展，其在多样化场景中的应用需求日益增长。然而，大语言模型的高计算资源需求与边缘设备有限的计算能力之间存在显著矛盾。小语言模型凭借其轻量级和高效的特点，为解决这一矛盾提供了潜在方案。本节分析由实际设备端需求驱动的大语言模型与小语言模型协作的典型应用场景。

4.1 实时低延迟推理场景

实时低延迟推理是边缘计算的核心需求，尤其是在需要即时响应的应用中，例如可穿戴设备/物联网语音助手和工业设备异常检测。LLM 与 SLM 的协作通过模型优化和边云协同框架，显著降低了推理延迟，提升了用户体验和系统效率。

可穿戴设备和物联网语音助手需要快速处理用户指令，以提供流畅的交互体验。LLM 强大的语言理解能力使其成为语音助手的理想选择，但其高计算需求需要通过 SLM 和边缘计算进行优化。Froiz-Miguez 等人的研究 (**froiz2023design**) 表明，依赖云端处理的传统语音助手

This involves different communication methods, such as passing text-based prompts and responses, sharing intermediate representations (like embeddings, hidden states), or using specialized APIs for inter-model communication (**king2024thoughtful**; **pozdniakov2024large**; **wozniak2024personalized**). Designing standardized interfaces that allow different models to interact effectively, regardless of their underlying architecture, is also an important consideration. Additionally, challenges related to ensuring data consistency and format compatibility between models need to be addressed. The efficiency and effectiveness of collaboration are directly influenced by the degree of communication and information exchange between models. Well-designed interfaces and communication protocols are essential for achieving seamless interaction. The choice of communication methods and interface design should be tailored to the specific collaboration mode and the type of information being exchanged. Pipeline collaboration might only require passing final outputs, whereas more tightly integrated models may benefit from sharing intermediate representations.

3.2.3 Model Fusion and Result Integration

Model fusion refers to merging the architectures or parameters of LLMs and SLMs to create a superior model, including techniques like weight averaging, knowledge fusion, ensemble learning, probability distribution-based fusion, model stacking, and mixture-of-experts models (**tang2024small**; **wan2024knowledge**; **yang2024model**; **mavromatis2024pack**; **wang2025speculate**; **shi2024profuser**).

Result integration refers to combining the outputs of collaborative models. Common methods include simple averaging, weighted averaging, majority voting, or using another model to combine the outputs. Additionally, handling potential inconsistencies or conflicts in the outputs of different models is necessary. Model fusion and result integration are crucial for creating unified systems that can leverage the distinct strengths and knowledge of multiple models. The choice of technique depends on the specific goals of the collaboration and the characteristics of the involved models. Evaluating the effectiveness of model fusion and result integration requires careful consideration of appropriate metrics and benchmarks to assess the performance of the combined system.

3.2.4 State Synchronization and Context Management

In collaborative LLMs and SLMs, especially in multi-turn dialogues or sequential tasks, maintaining a consistent state and managing context is crucial. This involves synchronizing the internal states (memory, attention weights) of the collaborating models and managing and sharing contextual information between them (**subramanian2025small**; **wang2024comprehensive**; **naveed2023comprehensive**; **mojarradi2024improving**), for example, using shared memory modules or passing context along with input. Additionally, challenges related to handling long context scenarios and ensuring contextual coherence across multiple interactions need to be addressed.

Maintaining state consistency and effectively managing context are essential for ensuring coherent and meaningful interactions in collaborative LLM-SLM systems, especially for tasks involving multiple steps or turns. The complexity of state synchronization and context management increases with the number of collaborating models and the length of interactions. Developing scalable and efficient techniques to handle these aspects is an ongoing research challenge.

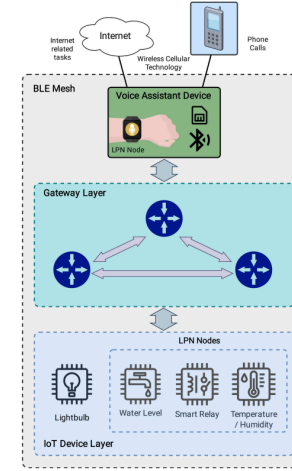


Figure 5: 边云协同 LLM-SLM 架构

在弱网络环境下用户体验会显著下降。为解决此问题，研究人员提出了边云协同的 LLM-SLM 架构，如图 5 所示。

Hao 等人提出的 Hybrid SLM-LLM 框架 (**hao2024hybrid**) 展示了一种动态令牌级边云协同推理方法，如图 6 所示。在该方法中，一个轻量级 SLM 部署在边缘设备上，用于处理基本语音指令并生成初步响应。复杂任务则基于不确定性量化机制，被路由到云端 LLM 进行进一步处理。该方法在延迟和准确性之间实现了最佳平衡，特别适用于资源受限的可穿戴设备场景。实验表明，与纯云端模式相比，该方法可将端到端延迟降低高达 40%，同时保持较高的响应质量。

在实际应用中，谷歌的 Gboard 和 SwiftKey 已采用类似方法，将本地 SLM 集成到移动键盘中用于语音识别和文本预测，仅在需要更复杂理解时才调用云服务 (**kumar2025large**)。这种协同方法显著提升了用户体验的流畅度和隐私保护水平。

工业场景中的设备监控与异常检测要求系统在毫秒级内响应，以防止潜在事故。研究表明，虽然 LLM 具备强大的分析能力，但单独部署难以满足工业环境的实时性要求 (**zheng2025review**)。

为应对这一挑战，研究人员提出了“推测解码”机制，利用边缘部署的 SLM 进行初步异常识别和风险评估，仅在检测到潜在风险时才激活更全面的 LLM 分析 (**friha2024llm**)。Zheng 等人的研究 (**zheng2025review**) 表明，这种协同模式可将工业异常检测的响应时间从秒级降低到毫秒级，同时保持与纯 LLM 方案相当的异常检测准确率。

WinCLIP 和 AnomalyGPT 等模型专门针对工业异常检测场景进行了优化。通过设备端 SLM 与云端 LLM 的协同，它们实现了准确性与实时性能的双重优化 (**zheng2025review**)。这一范式在化工生产、电力输送等高危工业场景中展现出显著价值。

3.2.5 Dynamic Resource Scheduling and Optimization

In collaborative LLM-SLM systems, especially when dealing with dynamic workloads and changing resource demands, efficiently allocating and managing computational resources (such as GPU memory, processing time) is a significant challenge (wang2024comprehensive).

Dynamic resource scheduling techniques can adjust resource allocation based on the current needs of the collaborating models and available resources. Optimization strategies, such as quantization, pruning, and efficient attention mechanisms, can be used to reduce the computational cost and memory footprint of collaborative inference (huang2025dynamic; amayuelas2025self; sun2024llumnix). The combination of cloud computing and edge computing also offers potential for resource management, for example, LLMs can run in the cloud while SLMs run on edge devices (lv2025collaboration). Efficient resource scheduling and optimization are essential for making collaborative LLM-SLM systems practical and cost-effective, especially in resource-constrained or high-traffic real-world applications. The heterogeneity of LLMs and SLMs in terms of resource requirements and performance characteristics adds complexity to resource scheduling and optimization. Policies need to consider these differences to achieve optimal resource utilization.

4 Application Scenarios Driven by On-Device Needs

With the rapid development of Large Language Models (LLMs), the demand for their application in diverse scenarios is growing. However, the high computational resource requirements of LLMs conflict significantly with the limited computing capabilities of edge devices. Small Language Models (SLMs), due to their lightweight and efficient characteristics, offer a potential solution to this contradiction. This section analyzes typical application scenarios where LLMs and SLMs collaborate, driven by actual on-device needs.

4.1 Real-time Low-Latency Inference Scenarios

Real-time low-latency inference is a core requirement of edge computing, especially in applications needing immediate responses, such as wearable/IoT voice assistants and industrial equipment anomaly detection. The collaboration between LLMs and SLMs significantly reduces inference latency and improves user experience and system efficiency through model optimization and edge-cloud collaborative frameworks.

Wearable devices and IoT voice assistants require rapid processing of user commands to provide a smooth interactive experience. The powerful language understanding capabilities of LLMs make them ideal for voice assistants, but their high computational demands necessitate optimization through SLMs and edge computing. Research by Froiz-Miguez et al. (froiz2023design) shows that traditional voice assistants relying on cloud processing suffer significant user experience degradation in weak network environments. To address this, researchers have proposed an edge-cloud collaborative LLM-SLM architecture, as shown in Figure 5.

The Hybrid SLM-LLM framework proposed by Hao et al. (hao2024hybrid) demonstrates a dynamic token-level edge-cloud collaborative inference method, depicted in Figure 6. In this approach, a lightweight SLM is deployed on the edge device to handle basic voice commands and generate preliminary responses. Complex tasks are routed to the cloud LLM for further processing based on an uncertainty quantification mechanism. This method achieves an optimal balance between latency and accuracy, particularly suitable for resource-constrained wearable device scenarios. Experiments show that compared to a purely cloud-based mode,

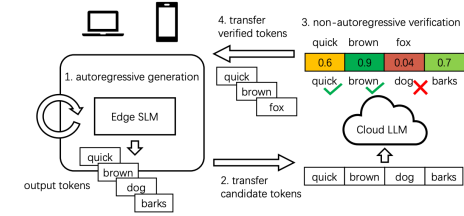


Figure 6: Hybrid SLM-LLM 框架结构

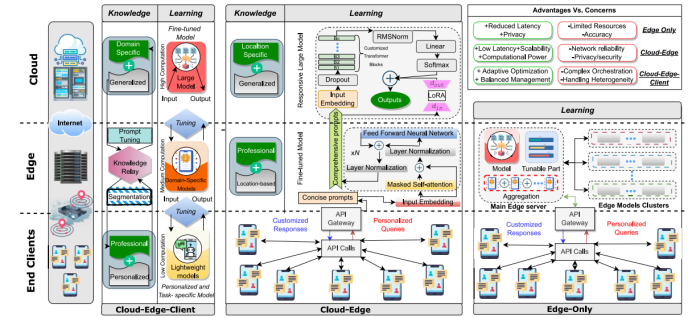


Figure 7: LLM 边缘部署策略

4.2 隐私敏感与本地数据场景

在涉及敏感数据的场景中，例如移动端个人知识库检索和医学影像预筛查，本地数据处理是保护隐私的关键。LLM 与 SLM 的协作通过本地部署和高效推理，满足了隐私保护和数据安全的需求。

移动设备上的个人知识库检索与问答系统需要快速响应用户查询，同时确保数据不离开设备以保护隐私。个人知识库通常包含敏感信息，直接将其上传至云端供 LLM 处理存在隐私泄露风险。为解决此问题，研究者提出了隐私保护的本地知识库检索框架。Chen 等人 (chen2024llm) 指出，在移动设备上部署 SLM 可以有效处理包含敏感信息的本地数据，而无需将原始数据传输至云端。

一个典型的实现是“隐私敏感检索增强生成”（Privacy-Sensitive RAG）架构。在此架构中，本地 SLM 负责敏感数据处理和初步检索，仅将去标识化的查询发送至云端 LLM 进行深度分析 (wang2024comprehensive)。这种方法在保护用户隐私的同时，仍能充分利用 LLM 强大的语义理解能力。

FedCoLLM (fan2024fedcollm) 进一步提出了一个联邦学习框架，使移动设备上的 SLM

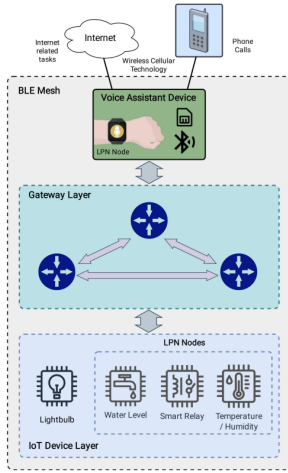


Figure 5: Edge-cloud collaborative LLM-SLM architecture

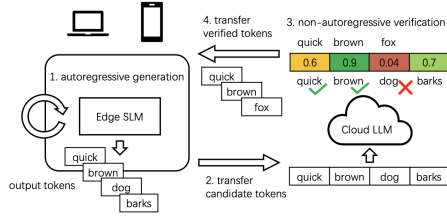


Figure 6: Hybrid SLM-LLM framework structure

this method can reduce end-to-end latency by up to 40% while maintaining high response quality.

In practical applications, Google's Gboard and SwiftKey have adopted similar methods, integrating local SLMs into mobile keyboards for speech recognition and text prediction, only calling cloud services when more complex understanding is needed (kumar2025large). This collaborative approach significantly enhances the smoothness of user experience and privacy protection levels.

Device monitoring and anomaly detection in industrial scenarios require systems to respond within milliseconds to prevent potential accidents. Research indicates that while LLMs have strong analytical capabilities, deploying them alone struggles to meet the real-time requirements of industrial settings (zheng2025review).

To tackle this challenge, researchers have proposed the "Speculative Decoding" mechanism, using edge-deployed SLMs for initial anomaly identification and risk assessment, only activating more comprehensive LLM analysis when potential risks are detected (friha2024llm).

能够从云端 LLM 获取知识，同时保护用户隐私数据。该框架通过参数高效的联合调优机制增强了设备端 SLM 的能力，为移动端隐私保护知识库提供了新途径。

医疗数据具有高度敏感性。将数据发送至云端进行分析的传统做法不仅存在隐私风险，还可能违反相关法规。研究表明，在本地部署 SLM 可以有效解决此问题 (niu2025collaborative)。

Zheng 等人 (zheng2025review) 提出了一个“分层诊断”模型，将初步筛查和基本诊断任务分配给本地部署的 SLM，仅在遇到复杂病例或需要专家意见时才调用云端 LLM。实验表明，该方法在诊断常见疾病方面表现良好，同时显著减少了敏感医疗数据的传输。

诸如 BioMistral (lan2019albert) 和 PathChat (lu2024multimodal) 等医学领域专用模型采用了类似的架构。设备端 SLM 处理基础医疗数据并进行初步筛查，而云端 LLM 则提供更专业的诊断建议。这种协作模式在保护患者隐私的同时，也提高了医疗服务的可及性，特别是在偏远地区和资源有限的医疗机构中。

4.3 任务特定定制与个性化场景

通过微调与适配器技术，LLM 和 SLM 能够针对特定任务进行定制，满足个人与企业的个性化需求，例如个人写作助手和企业内部知识管理。

个性化写作助手需要深入理解用户的长期写作风格与偏好，并具备实时响应能力 (liu2024contemporary)。传统的基于云的 LLM 解决方案虽然强大，但缺乏对个人数据的长期记忆和快速适应能力。研究者提出了一种“混合记忆架构”，将用户的写作风格特征、词汇偏好和常用表达保存在本地部署的 SLM 中，形成个性化的写作档案 (deng2023mutual)。当用户需要更深层次的创意或复杂内容生成时，系统调用云端 LLM 进行支持，并将交互结果反馈给本地 SLM，以持续优化个性化服务。

Liu 等人的实验 (liu2024contemporary) 表明，与纯云端或纯本地解决方案相比，这种协作模式将写作建议的个性化程度提高了高达 35%，同时将响应时间减少了 70%。主流文档编辑工具如 Microsoft Word 和 Google Docs 已开始采用类似架构，通过本地模型提供即时写作建议，并将复杂任务委托给云端处理。

企业知识管理系统需要处理大量专业领域知识，同时确保信息安全与快速访问。Jin 等人提出的 CE-CoLLM 框架 (jin2024collm) 为企业场景设计了一种云边协同的 LLM 推理架构，支持低延迟的边缘独立推理和高精度的云边协同推理两种模式。

在此框架下，企业可以在内部服务器上部署 SLM 来处理常见查询和基本任务，同时通过“提前退出机制”和“云端上下文管理器”在必要时高效调用外部 LLM 资源 (jin2024collm)。这种方法在确保敏感信息安全的同时，满足了高性能需求。

软银与 Aizip 合作开发的企业级定制 SLM 解决方案进一步证明了该方法的实用价值 (Aizip2024Softbank)。利用领域特定数据微调的 SLM 部署在企业环境中，能够处理企业特有的任务，其准确性与体积大 100 倍的云端 LLM 相当，同时显著降低了延迟和成本 (Aizip2024Softbank)。

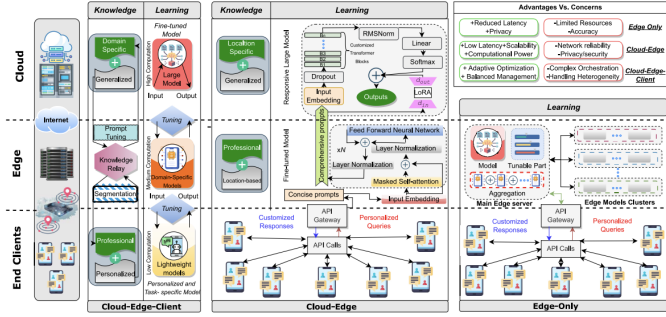


Figure 7: LLM Edge Deployment Strategy

Research by Zheng et al. (zheng2025review) shows that this collaborative mode can reduce response time in industrial anomaly detection from seconds to milliseconds, while maintaining anomaly detection accuracy comparable to pure LLM solutions.

Models like WinCLIP and AnomalyGPT are specifically optimized for industrial anomaly detection scenarios. Through the collaboration of on-device SLMs and cloud-based LLMs, they achieve dual optimization of accuracy and real-time performance (zheng2025review). This paradigm demonstrates significant value in high-risk industrial scenarios such as chemical production and power transmission.

4.2 Privacy-Sensitive and Local Data Scenarios

In scenarios involving sensitive data, such as personal knowledge base retrieval on mobile terminals and medical image pre-screening, local data processing is key to protecting privacy. The collaboration of LLMs and SLMs meets the needs for privacy protection and data security through local deployment and efficient inference.

Personal knowledge base retrieval and Q&A systems on mobile devices need to respond quickly to user queries while ensuring data does not leave the device to protect privacy. Personal knowledge bases often contain sensitive information, and uploading them directly to the cloud for LLM processing carries privacy leakage risks. To solve this, researchers proposed privacy-preserving local knowledge base retrieval frameworks. Chen et al. (chen2024llm) pointed out that deploying SLMs on mobile devices can effectively handle local data containing sensitive information without transmitting raw data to the cloud.

A typical implementation is the "Privacy-Sensitive Retrieval-Augmented Generation" (Privacy-Sensitive RAG) architecture. In this architecture, a local SLM is responsible for sensitive data processing and preliminary retrieval, sending only de-identified queries to the cloud LLM for in-depth analysis (wang2024comprehensive). This method protects user privacy while still fully utilizing the powerful semantic understanding capabilities of LLMs.

FedCoLLM (fan2024fedcollm) further proposed a federated learning framework enabling SLMs on mobile devices to acquire knowledge from cloud LLMs while protecting user privacy data. This framework enhances the capabilities of on-device SLMs through parameter-efficient joint tuning mechanisms, providing a new approach for privacy-preserving knowledge bases on mobile terminals.

4.4 离线或弱网络环境场景

在网络不可用或网络连接不稳定的环境中，LLM 和 SLM 的本地运行作为关键任务提供了重要支持，适用于远洋船舶、森林巡检无人机和灾害现场应急指挥等场景。

远洋船舶和森林巡检无人机通常在网络覆盖有限或没有网络的环境中运行，传统的依赖云端的 LLM 在此类场景下无法有效工作。为解决这一问题，研究人员提出了一种“离线优先”的 SLM-LLM 协同架构。实践表明，经过领域知识微调的 SLM 可以完全离线执行大多数基本交互任务。Kalita (kalita2024large) 的研究发现，通过对特定领域数据进行充分微调，拥有 40-50 亿参数的 SLM 在离线环境中可以处理 70-80% 的常见查询，其性能接近在线 LLM。这些系统在网络连接恢复时与云端 LLM 同步，更新本地模型的知识库和参数 (kalita2024large)。

一个典型应用是用于森林巡检的无人机系统。通过为无人机配备经过林业知识微调的 SLM，它们可以在离线状态下识别和分析植被状况、火灾风险等关键信息，显著提升巡检效率和可靠性 (lee2025multi)。

在灾害响应、边境救援等紧急情况下，网络基础设施可能遭到破坏，决策支持系统的实时性直接影响救援效果。Tong 等人 (tong2024robots) 的研究表明，传统的云端 LLM 在此类场景下缺乏可靠性，而纯本地模型则往往因能力有限而无法处理复杂情况。

为解决此问题，研究人员开发了一种“弹性协作”框架。在网络正常情况下，本地 SLM 与云端 LLM 协同工作，形成完整的决策支持链。当网络中断时，本地 SLM 可以利用预缓存的知识独立运行，确保基本的决策支持能力 (boateng2025survey)。

一个成功的案例是应用于地震救援的多智能体协同系统。该系统集成了离线部署的小语言模型和远程大语言模型，使得救援人员即使在网络中断期间也能获得基本的决策支持，并在网络恢复后立即获得更全面的分析 (Jayant2024StateEdgeAI_misc)。实验表明，与单一模型解决方案相比，这种弹性协作解决方案在极端环境下能提供显著更优的决策支持能力。

4.5 能量受限与绿色人工智能场景

在能量受限场景中，SLM 和优化后的 LLM 通过低功耗设计满足边缘设备的需求，例如超低功耗终端上的键盘预测和车载系统中的驾驶安全提示。

在具有严格功耗限制的终端设备上，如智能手表和功能受限的物联网设备，传统的云端 LLM 模式由于需要持续的网络连接，会显著缩短电池续航。研究表明，像键盘预测这样的功能，虽然不需要 LLM 级别的理解能力，但要求极低的延迟和能耗 (qu2025mobile)。

Kumar 的研究 (kumar2025large) 介绍了一种专为键盘预测优化的超低功耗 SLM 设计。通过极端的模型压缩和针对硬件的优化，预测功能的功耗降低到了传统方法的十分之一。尽管这些模型仅有数百万参数，但在特定场景下表现良好。

与 GloVe 合作开发的移动键盘预测 SLM 进一步证明了 SLM 在此场景中的价值 (kumar2025large)。该系统仅在复杂语境或用户明确请求时才唤醒更强大但能耗更高的模型，在预测准确性和能耗之间实现了最佳平衡，从而延长了设备使用时间。

车载系统面临着独特的能耗和实时性挑战。过高的功耗可能影响车辆整体电气系统，而与安全相关的提示则需要毫秒级的响应 (lin2023tiny)。传统的 LLM 模式难以满足这些需求。

Qu 等人 (qu2025mobile) 提出了一种“分层安全提示”架构。通过在车载系统内部署

Medical data is highly sensitive. The traditional practice of sending data to the cloud for analysis not only poses privacy risks but may also violate relevant regulations. Research shows that deploying SLMs locally can effectively address this issue (**niu2025collaborative**).

Zheng et al. (**zheng2025review**) proposed a "hierarchical diagnosis" model, assigning preliminary screening and basic diagnostic tasks to locally deployed SLMs, only invoking cloud LLMs when encountering complex cases or requiring expert opinions. Experiments demonstrate that this method performs well in diagnosing common diseases while significantly reducing the transmission of sensitive medical data.

Medical domain-specific models like BioMistral (**lan2019albert**) and PathChat (**lu2024multimodal**) adopt similar architectures. On-device SLMs process basic medical data and perform initial screening, while cloud LLMs provide more specialized diagnostic advice. This collaborative mode protects patient privacy while also improving the accessibility of medical services, especially in remote areas and resource-limited medical institutions.

4.3 Task-Specific Customization and Personalization Scenarios

Through fine-tuning and adapter techniques, LLMs and SLMs can be customized for specific tasks, meeting the personalized needs of individuals and enterprises, such as personal writing assistants and internal corporate knowledge management.

Personalized writing assistants require a deep understanding of the user's long-term writing style and preferences, along with real-time responsiveness (**liu2024contemporary**). Traditional cloud-based LLM solutions, while powerful, lack long-term memory of personal data and rapid adaptation capabilities. Researchers proposed a "Hybrid Memory Architecture," maintaining user writing style features, vocabulary preferences, and common expressions in a locally deployed SLM, forming a personalized writing archive (**deng2023mutual**). When the user needs deeper creativity or complex content generation, the system calls the cloud LLM for support and feeds the interaction results back to the local SLM to continuously optimize personalized services.

Experiments by Liu et al. (**liu2024contemporary**) show that this collaborative mode increases the personalization of writing suggestions by up to 35% compared to pure cloud or pure local solutions, while reducing response time by 70%. Mainstream document editing tools like Microsoft Word and Google Docs have begun adopting similar architectures, providing instant writing suggestions via local models and delegating complex tasks to the cloud.

Enterprise knowledge management systems need to handle large amounts of specialized domain knowledge while ensuring information security and rapid access. The CE-CoLLM framework proposed by Jin et al. (**jin2024collm**) designs a cloud-edge collaborative LLM inference architecture for enterprise scenarios, supporting both low-latency edge-independent inference and high-precision cloud-edge collaborative inference modes.

Under this framework, enterprises can deploy SLMs on internal servers to handle common queries and basic tasks, while efficiently invoking external LLM resources when necessary through an "early-exit mechanism" and a "cloud context manager" (**jin2024collm**). This approach ensures sensitive information security while meeting high-performance requirements.

The enterprise-grade customized SLM solution developed through collaboration between SoftBank and Aizip further proves the value of this approach (**Aizip2024Softbank**). SLMs fine-tuned with domain-specific data are deployed within the enterprise environment, capable of handling unique enterprise tasks with accuracy comparable to cloud LLMs 100 times larger, while significantly reducing latency and costs (**Aizip2024Softbank**).

一个专门掌握安全驾驶知识的 SLM 进行实时环境监控和基本安全提示,同时保留连接云端 LLM 以进行复杂情况分析的能力,该架构在保持安全提示准确性的同时,降低了 65% 的能耗。

一些汽车制造商已开始采用类似的“绿色人工智能”解决方案,使用本地 SLM 处理导航提示、语音控制和基本安全警告,仅在需要深度驾驶辅助时才激活更强大的模型 (**qu2025mobile**)。这种方法不仅显著降低了能耗,还提高了系统可靠性,尤其是在网络连接不稳定的区域。

5 挑战与开放性问题

大型语言模型 (LLM) 与小型语言模型 (SLM) 的协作旨在结合两者的优势: LLM 强大的通用能力和知识广度,以及 SLM 在特定任务上的高效性、低延迟和低成本。然而,实现完美的协作并非易事,面临着诸多挑战和尚未解决的开放性问题。本节将深入探讨这些挑战,主要聚焦于协作效率与系统开销、模型间一致性与兼容性、任务分配策略的鲁棒性与最优性、协作系统的训练与维护、评估指标与基准测试,以及相关的安全、隐私和伦理问题。

5.1 协作效率与系统开销

5.1.1 任务路由决策的延迟

尽管大语言模型与小语言模型的协作旨在提升整体效率,但协作过程本身可能引入新的开销,从而对系统性能产生潜在的负面影响。关键挑战在于任务路由决策的延迟、模型间通信的成本以及整体系统的复杂性管理。

在大语言模型与小语言模型的协作系统中,任务路由机制负责决定应由哪个模型处理特定的查询或子任务,这是实现协作效率的核心。然而,路由决策过程本身会引入延迟,可能部分抵消使用小语言模型所带来的速度优势 (**arcee2025slms**)。挑战在于设计既准确又快速的路由机制。

小语言模型因其低推理延迟而备受青睐,尤其适用于边缘设备。相比之下,大语言模型由于其庞大的参数量和高计算需求,通常具有更高的推理延迟 (**wang2024comprehensive**)。协作系统需要一个路由机制,以便根据任务复杂性或其他标准动态选择模型 (**arize2025survey**)。这些路由决策,无论是基于不确定性估计、词元级分析、任务分解还是其他预测方法,都会产生其自身的计算开销和延迟。级联方法(先查询小语言模型,必要时再升级到大型语言模型)会明确引入延迟和潜在的冗余。像 CITER 这样的框架试图通过词元级路由来最小化对大语言模型的调用,但仍需要一个路由机制 (**zheng2025citer**)。预生成路由旨在通过预先推断大语言模型的能力来最小化延迟,但这种预测本身可能很复杂 (**varangot2025doing**)。

存在一个核心悖论:使用小语言模型的主要动机通常是减少延迟,然而,有效选择何时使用小语言模型(即路由器)的机制本身却引入了延迟。路由器的延迟成为一个关键因素;如果它太高,那么协作带来的净延迟减少(小语言模型延迟 + 路由器延迟对比大语言模型延迟)就会减弱。这提出了一个开放的研究挑战:开发近乎零延迟的路由机制,或者证明路由开销与大语言模型-小语言模型之间的延迟差异相比始终可以忽略不计。* 开放性问题: * 开发超

4.4 Offline or Weak Network Environment Scenarios

In environments with no network or unstable network connectivity, the local operation of LLMs and SLMs provides crucial support, applicable to scenarios like ocean-going vessels, forest inspection drones, and emergency command at disaster sites.

Ocean-going vessels and forest inspection drones often operate in environments with limited or no network coverage, where traditional cloud-reliant LLMs are ineffective. To address this, researchers proposed an "offline-first" SLM-LLM collaborative architecture. Practice shows that SLMs fine-tuned with domain knowledge can perform most basic interaction tasks completely offline. Research by Kalita (**kalita2024large**) found that through sufficient fine-tuning on specific domain data, SLMs with 4-5 billion parameters can handle 70-80% of common queries in offline environments, with performance approaching that of online LLMs. These systems synchronize with cloud LLMs when network connectivity is restored, updating the local model's knowledge base and parameters (**kalita2024large**).

A typical application is a drone system for forest inspection. By equipping drones with SLMs fine-tuned with forestry knowledge, they can identify and analyze vegetation conditions, fire risks, and other key information offline, significantly improving inspection efficiency and reliability (**lee2025multi**).

In emergency situations like disaster response and border rescue, network infrastructure may be damaged, and the real-time performance of decision support systems directly impacts rescue effectiveness. Research by Tong et al. (**tong2024robots**) indicates that traditional cloud LLMs lack reliability in such scenarios, while purely local models are often incapable of handling complex situations due to limited capabilities.

To address this, researchers developed a "Resilient Collaboration" framework. Under normal network conditions, the local SLM collaborates with the cloud LLM, forming a complete decision support chain. When the network is interrupted, the local SLM can operate independently using pre-cached knowledge, ensuring basic decision support capabilities (**boateng2025survey**).

A successful case is a multi-agent collaborative system used in earthquake rescue. This system integrates offline-deployed small language models and remote large language models, enabling rescue personnel to receive basic decision support even during network outages, and immediately access more comprehensive analysis once the network recovers (**Jayant2024StateEdgeAI_misc**). Experiments show that this resilient collaboration solution offers significantly better decision support capabilities in extreme environments compared to single-model solutions.

4.5 Energy-Constrained and Green AI Scenarios

In energy-constrained scenarios, SLMs and optimized LLMs meet the needs of edge devices through low-power design, such as keyboard prediction on ultra-low power terminals and driving safety prompts in in-vehicle systems.

On terminal devices with strict power constraints, like smartwatches and limited-functionality IoT devices, the traditional cloud LLM mode significantly reduces battery life due to continuous network connection requirements. Research shows that functions like keyboard prediction, while not requiring LLM-level understanding, demand extremely low latency and energy consumption (**qu2025mobile**).

Research by Kumar (**kumar2025large**) introduced an ultra-low power SLM design specifically optimized for keyboard prediction. Through extreme model compression and hardware-specific optimization, the power consumption for prediction functions was reduced to 1/10th of traditional methods. Although these models have only millions of parameters, they perform well in specific scenarios.

低延迟路由机制；量化路由复杂性/准确性与延迟之间的权衡；设计能够通过并行处理掩盖路由延迟的系统。

5.1.2 模型间通信成本

在协作系统中，尤其是在边云协同场景中（SLM 本地运行，LLM 在云端运行），模型之间的数据传输（例如查询、中间结果、上下文、模型输出）会产生通信成本和延迟。这对于实时应用程序或处理大量数据时尤其具有挑战性。

云边协作是一种常见的模式，旨在平衡隐私/延迟（设备端 SLM）与能力（云端 LLM）。将用户数据实时传输到云端会引发对通信成本、延迟和带宽限制的担忧。像 LSC4Rec 这样的框架旨在通过让 LLM 仅向设备端小型推荐模型（SRM）发送候选列表进行重新排序，以最小化实时数据传输 (**lv2025collaboration**)。类似地，MinionS 在云端分解任务，将子任务发送到本地 SLM，并在云端聚合结果，旨在减少与云端交互的数据量和频率 (**narayan2025minions**)。然而，通信开销仍然是一个因素，尤其是在需要交换中间结果或大量上下文时。

虽然边云协作利用了本地 SLM 的效率和云端 LLM 的强大能力，但通信通道本身可能成为瓶颈，限制系统的整体响应能力和成本效益。协作协议的设计和任务分解的性质对于管理通信开销至关重要。优化通信内容和时序是一个开放的挑战。* 开放性问题：* 设计通信高效的协作协议；开发用于压缩模型间交换的中间表示的技术；量化网络条件对协作系统性能的影响。

5.2 模型间一致性与兼容性

5.2.1 确保不同模型间输出风格与知识范围的一致性

大型语言模型（LLM）与小型语言模型（SLM）通常在不同数据集上训练或为不同目的进行微调，这可能导致它们展现出不同的写作风格、语调或响应格式。当不同模型为同一输出或对话做出贡献时，确保一致的用户体验需要机制来对齐或管理这些风格差异。

一致性被认为是 LLM 可靠性的一个基本方面。诸如"指导链"（CoG）等技术旨在通过引导式提示来改进 LLM 输出的语义一致性 (**raj2025improving**)。像 SAG 这样的框架，通过协同训练（S-SFT, C-DPO）和自我改进方法，明确解决了 LLM-SLM 协作中的风格对齐问题 (**xusag**)。模块化多元主义框架使用较小的"社区 LM"将不同的视角/风格整合到基础 LLM 中 (**feng2024modular**)。评估一致性本身具有挑战性，通常需要超越简单相似度的细致指标 (**raj2025improving**)。KnowsLM 框架使用 LLM 作为评判者来评估风格适应能力，发现微调在语调适应方面优于 RAG (**harbola2025knowslm**)。

协作本身存在风格冲突的风险。在 CoGenesis 框架中，如果 SLM 处理个人细节而 LLM 处理更广泛的叙述 (**zhang2024cogenesis**)，除非采用特定的对齐技术，否则组合输出可能会显得脱节。这需要在训练/微调过程中标准化风格，或开发复杂的输出融合技术。由于 LLM 和 SLM 通常具有不同的训练历史和目标，这导致了潜在不同的输出风格、语调和格式。在协作系统中，不同模型可能生成最终输出的各个部分。将风格迥异的模型输出组合起来，可能导致不一致且令人不适的用户体验。确保一致性需要明确的努力，要么通过联合训练/微调以实现风格对齐 (**xusag**)，要么通过后处理/融合技术 (**pei2025mathfusion**)。这增加了系统设计

The mobile keyboard prediction SLM developed in collaboration with GloVe further demonstrates the value of SLMs in this scenario (**kumar2025large**). The system only wakes up more powerful but energy-intensive models in complex contexts or when explicitly requested by the user, achieving an optimal balance between prediction accuracy and energy consumption, thereby extending device usage time.

In-vehicle systems face unique energy consumption and real-time challenges. Excessive power consumption can affect the vehicle's overall electrical system, while safety-related prompts require millisecond-level responses (**lin2023tiny**). Traditional LLM modes struggle to meet these demands.

Qu et al. (**qu2025mobile**) proposed a "Hierarchical Safety Prompt" architecture. By deploying an SLM specialized in safe driving knowledge within the vehicle system for real-time environment monitoring and basic safety prompts, while retaining connectivity to a cloud LLM for complex situational analysis, this architecture reduces energy consumption by 65% while maintaining the accuracy of safety prompts.

Several automakers have begun adopting similar "Green AI" solutions, using local SLMs to handle navigation prompts, voice control, and basic safety warnings, only activating more powerful models when deep driving assistance is needed (**qu2025mobile**). This approach not only significantly reduces energy consumption but also improves system reliability, especially in areas with unstable network connections.

5 Challenges and Open Issues

The collaboration between Large Language Models (LLMs) and Small Language Models (SLMs) aims to combine the strengths of both: the powerful general capabilities and knowledge scope of LLMs, and the efficiency, low latency, and low cost of SLMs on specific tasks. However, achieving perfect collaboration is not straightforward and faces numerous challenges and unresolved open questions. This section delves into these challenges, focusing primarily on collaboration efficiency and system overhead, inter-model consistency and compatibility, robustness and optimality of task allocation strategies, training and maintenance of collaborative systems, evaluation metrics and benchmarking, and related security, privacy, and ethical issues.

5.1 Collaboration Efficiency and System Overhead

5.1.1 Latency of Task Routing Decisions

Although LLM-SLM collaboration aims to enhance overall efficiency, the collaboration process itself can introduce new overheads, potentially negatively impacting system performance. Key challenges lie in the latency of task routing decisions, the cost of inter-model communication, and the complexity management of the overall system.

In LLM-SLM collaborative systems, the task routing mechanism is responsible for determining which model should handle a specific query or sub-task, which is core to achieving collaborative efficiency. However, the routing decision process itself introduces latency, potentially partially offsetting the speed advantage gained from using SLMs (**arcee2025slms**). The challenge is to design routing mechanisms that are both accurate and fast.

SLMs are favored for their low inference latency, especially suitable for edge devices. In contrast, LLMs, due to their massive parameter count and high computational demands, typically have higher inference latency (**wang2024comprehensive**). Collaborative systems require a routing mechanism to dynamically select models based on task complexity or other criteria (**arize2025survey**). These routing decisions, whether based on uncertainty estima-

和评估的复杂性。* 开放性问题：* 开发用于评估不同模型间风格一致性的稳健指标；创建用于协作生成中实时风格对齐或迁移的高效技术；理解风格一致性与事实准确性或创造性等其他目标之间的权衡。

5.2.2 模型知识范围对齐与事实一致性

由于训练数据和模型能力的差异，大语言模型（LLM）和小语言模型（SLM）拥有不同的知识库。大语言模型通常知识面更广，但可能产生幻觉或包含过时信息；小语言模型可能拥有更专业、可能更及时的领域知识，但广度不足。确保模型协作时的事实一致性并解决知识冲突是一个重大障碍。

大语言模型存在幻觉和知识陈旧问题，而小语言模型在缺乏足够领域特定知识的专业领域往往表现不佳。知识蒸馏(KD)等技术旨在将知识从大语言模型迁移到小语言模型 (**wang2024comprehensive**) 但对齐复杂分布是困难的 (**peng2024enhancing**)。检索增强生成(RAG)被用于注入外部、最新的知识，但持续整合检索到的知识同样具有挑战性。使用 KnowsLM 框架的研究表明，RAG 擅长实时知识注入，而微调在风格一致性方面表现更佳 (**harbola2025knowslm**)。CoVer 框架利用小语言模型验证来检查大语言模型推理的一致性 (**yan2025collaborative**)。CrossLM 框架利用小语言模型的反馈来改进大语言模型生成的合成数据，以实现相互增强 (**deng2023mutual**)。

当大语言模型和小语言模型拥有相互冲突的信息时（由于不同的训练截止时间或专业领域），协作系统需要一种机制来解决这种冲突。简单的融合可能导致事实错误。这需要一个复杂的真相发现机制，或者一种基于特定查询类型明确信任某个模型知识领域的策略。大语言模型和小语言模型拥有不同的知识库；大语言模型可能产生幻觉或信息过时，而小语言模型可能缺乏广泛的知识或领域细节。协作可能导致模型提供相互冲突的事实信息。系统需要一种策略来处理这些冲突：根据上下文/任务优先考虑某个模型、使用外部验证（如 RAG）、或采用权衡置信度/时效性的融合机制。未能解决冲突将导致输出不可靠且不可信。

这需要在 LLM-SLM 协作系统中研究鲁棒的知识融合和冲突解决机制。* 开放性问题：* 为 LLM-SLM 输出开发有效的知识融合技术；创建基于来源可靠性或上下文的动态冲突解决机制；确保协作输出中信息来源的可追溯性；对齐协作模型间的知识更新。

全面评估 LLM-SLM 协作系统的性能是复杂的，不仅需要衡量最终结果的质量，还需要考虑效率、成本等多个维度。目前，该领域缺乏标准化的测试平台和数据集，这使得比较不同协作策略和衡量进展变得困难。

评估 LLM-SLM 协作系统不能仅仅依赖传统的 NLP 指标；还必须全面考虑协作带来的效率提升和成本节约。这需要一个能够捕捉协作系统多方面性能的评估框架。

LLMOps 的挑战超越了部署单一模型，还包括数据处理、模型训练、部署、维护、应对模型漂移以及确保适应不断变化的数据和任务。对于协作系统而言，这些挑战更为严峻。从业者认为模型部署和监控阶段既重要又困难。模型通常作为独立服务部署，MLOps 原则的采用有限。报告的问题包括为生产部署设计基础设施架构的困难，以及与遗留应用程序的集成问题。生产中的许多模型并未受到监控；关键的监控方面包括输入、输出和决策。挑战还包括缺乏监控实践、需要创建自定义监控工具以及选择合适的指标 (**zimelewicz2024ml**)。Acxiom 在使用 LLM 和 LangChain 创建受众细分系统时，在调试复杂 workflow 方面遇到了挑

tion, token-level analysis, task decomposition, or other predictive methods, incur their own computational overhead and latency. Cascading methods (querying SLM first, then escalating to LLM if needed) explicitly introduce latency and potential redundancy. Frameworks like CITER attempt to minimize LLM calls through token-level routing but still require a routing mechanism (**zheng2025citer**). Pre-generation routing aims to minimize latency by inferring LLM capabilities beforehand, but this prediction itself can be complex (**varangot2025doing**).

A core paradox exists: the primary motivation for using SLMs is often to reduce latency, yet the mechanism for effectively choosing when to use an SLM (i.e., the router) itself introduces latency. The router's latency becomes a critical factor; if it is too high, the net latency reduction from collaboration (SLM latency + router latency vs. LLM latency) diminishes. This presents an open research challenge: developing near-zero-latency routing mechanisms or demonstrating that routing overhead is consistently negligible compared to the LLM-SLM latency difference. ****Open Issues:**** Develop ultra-low-latency routing mechanisms; quantify the trade-offs between routing complexity/accuracy and latency; design systems capable of masking routing latency through parallel processing.

5.1.2 Cost of Inter-Model Communication

In collaborative systems, especially in edge-cloud scenarios (SLM runs locally, LLM in the cloud), data transfer between models (e.g., queries, intermediate results, context, model outputs) incurs communication costs and latency. This is particularly challenging for real-time applications or when handling large amounts of data.

Cloud-edge collaboration is a common pattern to balance privacy/latency (on-device SLM) with capability (cloud LLM). Transmitting user data in real-time to the cloud raises concerns about communication costs, latency, and bandwidth limitations. Frameworks like LSC4Rec aim to minimize real-time data transfer by having the LLM send only a candidate list to the on-device small recommendation model (SRM) for re-ranking (**lv2025collaboration**). Similarly, MinionS decomposes tasks in the cloud, sends sub-tasks to local SLMs, and aggregates results in the cloud, aiming to reduce the volume and frequency of cloud interactions (**narayan2025minions**). However, communication overhead remains a factor, especially when intermediate results or substantial context need to be exchanged.

While edge-cloud collaboration leverages the efficiency of local SLMs and the power of cloud LLMs, the communication channel itself can become a bottleneck, limiting the overall responsiveness and cost-effectiveness of the system. The design of collaboration protocols and the nature of task decomposition are critical for managing communication overhead. Optimizing communication content and timing is an open challenge. ****Open Issues:**** Design communication-efficient collaboration protocols; develop techniques for compressing intermediate representations exchanged between models; quantify the impact of network conditions on collaborative system performance.

5.2 Inter-Model Consistency and Compatibility

5.2.1 Ensuring Consistency in Output Style and Knowledge Scope Across Different Models

LLMs and SLMs are often trained on different datasets or fine-tuned for different purposes, which can lead them to exhibit distinct writing styles, tones, or response formats. When different models contribute to the same output or dialogue, ensuring a consistent user experience requires mechanisms to align or manage these stylistic differences.

Consistency is considered a fundamental aspect of LLM trustworthiness. Techniques like "Chain of Guidance" (CoG) aim to improve the semantic consistency of LLM outputs through

战。通过 LangSmith 实现的观测性优化了令牌使用，并有效扩展了他们的混合模型部署 (**zenml2025llmops**)。持续监控 LLM 并主动检测异常和对抗行为对于确保其完整性至关重要。将 LLM 集成到 CI/CD 管道中会引发与计算成本、不准确、错误处理、偏见以及开发、部署、维护和伦理相关的问题相关的担忧 (**pahune2025transitioning**)。协作系统中组件数量的增加意味着监控复杂性的提升和更多潜在的故障点。每个模型 (LLM、SLM)、路由机制以及它们之间的通信链路都需要监控其性能、资源消耗、错误率和漂移情况。关联来自不同组件的日志和指标以诊断系统范围的问题可能极具挑战性。需要专门为这种分布式、异构模型环境设计的监控和调试工具。协作系统比单模型系统拥有更多组件 (LLM、SLM、路由器)。每个组件都是一个潜在的故障点，需要单独监控其健康状况、性能 (延迟、吞吐量)、资源使用情况和输出质量。此外，组件之间的交互也需要监控，例如数据流、API 调用成功率和通信延迟。当问题出现时 (例如，整体响应质量下降)，精确定位哪个组件或交互出现故障可能非常复杂，需要复杂的日志记录、追踪和根本原因分析能力。这显著增加了部署和运维的复杂性。**** 开放性问题: **** 为 LLM-SLM 协作系统开发标准化的部署模式和最佳实践; 创建能够跨多个模型跟踪请求流和性能指标的全面监控框架; 为协作系统设计自动化的异常检测和故障诊断工具; 研究如何在分布式模型部署中有效管理和更新各个组件。

5.3 安全、隐私与伦理问题

尽管具有潜在优势，LLM-SLM 协作系统也可能引入或放大与安全、隐私和伦理相关的风险。不同模型之间的数据流动、协作决策过程以及模型自身固有的缺陷，都可能是这些问题的来源。

5.3.1 不同模型间数据流动的隐私风险

当数据，尤其是包含敏感用户信息的数据，在本地 SLM 与云端 LLM (甚至在不同 LLM 之间) 流动时，隐私泄露的风险会增加。

由于资源限制，使用云端 API 调用 LLM 很常见，这本身就引发了隐私担忧。SLM 通常部署在边缘设备上以保护隐私 (**wang2024comprehensive**)。LLM 可能会从训练数据或交互中泄露个人可识别信息 (PII)。像 CoGenesis 这样的框架旨在通过让 SLM 在本地处理私有数据，从逻辑上防止隐私泄露 (**zhang2024cogenesis**)。CrossLM 框架涉及在私有客户端数据上训练的 SLM 来增强云端 LLM，这引发了关于知识转移与数据泄露的问题 (**deng2023mutual**)。联邦协作被认为是一种平衡隐私的策略。

虽然云边协作分布了数据处理，但它也创建了更多的数据传输或处理节点，如果设计不当，可能会增加隐私泄露的攻击面。LLM-SLM 协作通常涉及模型之间的数据流动，可能跨越信任边界 (例如，设备到云端)。LLM 如果处理敏感用户数据，会带来隐私风险。设备上的 SLM 通常用于通过将敏感数据保留在本地来缓解这些风险。然而，协作本身可能需要共享一些信息 (中间结果、上下文摘要)，这些信息可能会无意中泄露隐私细节。确保共享的信息被充分匿名化或泛化，同时仍对协作有用，是一个重大挑战。这需要精心设计数据流协议，并可能在协作框架内采用隐私增强技术 (PETs)。 **** 开放性问题: **** 为 LLM-SLM 协作开发隐私保护的数据共享协议; 形式化混合系统中的隐私保证; 研究隐私保护与协作性能之间的权衡;

guided prompting (**raj2025improving**). Frameworks like SAG explicitly address style alignment in LLM-SLM collaboration using co-training (S-SFT, C-DPO) and self-improvement methods (**xusag**). The Modular Pluralism framework uses smaller "community LMs" to integrate diverse perspectives/styles into a base LLM (**feng2024modular**). Evaluating consistency itself is challenging, often requiring nuanced metrics beyond simple similarity (**raj2025improving**). The KnowsLM framework uses an LLM as a judge to evaluate style adaptation, finding fine-tuning better than RAG for tone adaptation (**harbola2025knowslm**).

Collaboration inherently risks stylistic clashes. In the CoGenesis framework, if the SLM handles personal details while the LLM handles the broader narrative (**zhang2024cogenesis**), the combined output might feel disjointed unless specific alignment techniques are employed. This necessitates standardizing styles during training/fine-tuning or developing sophisticated output fusion techniques. Since LLMs and SLMs often have different training histories and objectives, this leads to potentially divergent output styles, tones, and formats. In collaborative systems, different models might generate parts of the final output. Combining outputs from stylistically different models can result in an inconsistent and jarring user experience. Ensuring consistency requires explicit effort, either through joint training/fine-tuning for style alignment (**xusag**), or through post-processing/fusion techniques (**pei2025mathfusion**). This adds complexity to system design and evaluation. ****Open Issues:**** Develop robust metrics for evaluating stylistic consistency across different models; create efficient techniques for real-time style alignment or transfer in collaborative generation; understand the trade-offs between stylistic consistency and other objectives like factual accuracy or creativity.

5.2.2 Alignment of Model Knowledge Scope and Factual Consistency

Due to differences in training data and model capacity, LLMs and SLMs possess different knowledge bases. LLMs typically have broader knowledge but may hallucinate or contain outdated information; SLMs may have more specialized, potentially up-to-date domain knowledge but lack breadth. Ensuring factual consistency and resolving knowledge conflicts when models collaborate is a significant hurdle.

LLMs suffer from hallucination and knowledge staleness issues, while SLMs often underperform in specialized domains without sufficient domain-specific knowledge. Techniques like Knowledge Distillation (KD) aim to transfer knowledge from LLMs to SLMs (**wang2024comprehensive**), but aligning complex distributions is difficult (**peng2024enhancing**). RAG is used to inject external, up-to-date knowledge, but consistently integrating retrieved knowledge is also challenging. Research with the KnowsLM framework shows RAG excels at real-time knowledge injection, while fine-tuning is better for stylistic consistency (**harbola2025knowslm**). The CoVer framework uses SLM verification to check the consistency of LLM reasoning (**yan2025collaborative**). The CrossLM framework uses feedback from SLMs to improve synthetic data generated by LLMs for mutual enhancement (**deng2023mutual**).

When LLMs and SLMs possess conflicting information (due to different training cutoffs or specializations), the collaborative system needs a mechanism to resolve this conflict. Simple fusion could lead to factual errors. This requires a sophisticated truth discovery mechanism or a strategy to explicitly trust one model's knowledge domain over another based on the specific query type. LLMs and SLMs have different knowledge bases; LLMs can hallucinate or be outdated, while SLMs may lack broad knowledge or domain specifics. Collaboration can lead to models providing conflicting factual information. The system needs a strategy to handle these conflicts: prioritizing one model based on context/task, using external verification (like RAG), or employing a fusion mechanism that weighs confidence/recency. Failure to resolve conflicts results in unreliable and untrustworthy outputs.

This necessitates research into robust knowledge fusion and conflict resolution mechanisms

审计数据流以识别潜在的泄露点。

5.3.2 协作系统中潜在的偏见放大或安全漏洞

单个大语言模型和小语言模型可能携带其训练数据中的偏见。当这些模型协作时，这些偏见可能会被放大，或者由于交互动态而产生新的、涌现的偏见。确保组合系统的公平性并减轻偏见比单个模型更为复杂。

大语言模型可能会延续并放大训练数据中存在的社会偏见。小语言模型如果在较小或多样性不足的数据集上训练，也可能表现出偏见。大语言模型之间的迭代学习可能会放大细微的偏见 (**ren2024bias**)。ViLBias 框架指出，结合不同模态（文本和图像）可以揭示单一模态中不明显的偏见，这表明模型协作可能产生类似的效果 (**raza2024vilbias**)。大语言模型路由器本身也可能存在漏洞；如果攻击者控制了路由，将查询导向有偏见的模型，就可能被利用来诱导产生有偏见的结果 (**lin2025life**)。

偏见在协作系统中可能会叠加。如果一个语言模型基于对用户人口统计数据的偏见理解来路由查询，而选中的大语言模型随后又生成带有其自身偏见的内容，那么最终输出可能会受到双重偏见的影响。交互本身也可能产生个体模型中不存在的新形式偏见。大语言模型和小语言模型都可能因其训练数据或架构而产生偏见。在协作系统中，多个模型共同参与最终输出或决策过程。如果不同模型具有不同的偏见，这些偏见可能会以不可预测的方式相互作用。一个模型的偏见输出可能成为另一个模型的偏见输入，从而可能放大初始偏见（例如，一个有偏见的小语言模型将查询路由到一个大语言模型，后者随后强化了该偏见）。减轻此类系统中的偏见不仅需要评估单个组件，还需要评估整个协作流程及其决策逻辑。**** 开放性问题：**** 开发检测和衡量大语言模型-小语言模型协作系统中偏见放大的方法；设计公平的路由和任务分配策略；创建用于消除多模型系统输出偏见的技术；理解不同的协作架构如何影响偏见传播。

大语言模型-小语言模型系统的复杂性和分布式特性可能会引入新的安全漏洞。攻击者可能针对通信信道、路由机制，或利用一个模型的弱点来危害整个系统。大语言模型容易受到各种攻击，如提示注入、越狱、数据投毒和个人身份信息泄露。研究表明，大语言模型路由器本身存在生命周期漏洞，包括对抗性攻击和后门，其中基于深度神经网络的路由器尤其脆弱。如果路由器被攻破，它可能恶意地将查询路由到不适当的模型或泄露数据。大语言模型和小语言模型之间的交互点（例如，API 调用、数据交换）成为潜在的攻击面。如果边缘设备上的小语言模型被攻破，它可能向云端大语言模型发送恶意输入，反之亦然。协作系统中的每个额外模型和通信链路都可能扩大攻击面。协作系统中"最薄弱环节"（安全性较低的 SLM 或路由器）的漏洞可能危及更强大的 LLM 或整个系统的完整性/安全性。LLM-SLM 系统由多个组件（LLM、SLM、路由器、通信通道）组成。每个组件和接口都是潜在的攻击点。路由器可能遭受攻击导致误路由（将所有查询发送至昂贵的 LLM，或发送至已遭破坏/无效的 SLM）。通信通道若不安全，则允许模型间交换的数据被拦截或篡改。遭破坏的 SLM 可能向 LLM 输送恶意数据（反之亦然），可能导致有害输出或系统被接管。确保协作系统的安全性需要采用整体性方法，考虑每个组件及其交互的安全性，这比保护单一整体模型更为复杂。

**** 开放性问题：**** 为 LLM-SLM 协作开发安全通信协议；设计鲁棒且抗攻击的路由机

in LLM-SLM collaborative systems. ****Open Issues:**** Develop effective knowledge fusion techniques for LLM-SLM outputs; create dynamic conflict resolution mechanisms based on source reliability or context; ensure traceability of information sources in collaborative outputs; align knowledge updates across collaborating models.

5.3 Evaluation Metrics and Benchmarking

Comprehensively evaluating the performance of LLM-SLM collaborative systems is complex, requiring measurement not only of the quality of the final results but also consideration of multiple dimensions like efficiency and cost. Currently, the field lacks standardized testing platforms and datasets, making comparison between different collaborative strategies and measurement of progress difficult.

Evaluating LLM-SLM collaborative systems cannot rely solely on traditional NLP metrics; it must also comprehensively consider the efficiency gains and cost savings brought by collaboration. This necessitates an evaluation framework capable of capturing the multifaceted performance of collaborative systems.

The challenges of LLMops extend beyond deploying single models to include data processing, model training, deployment, maintenance, addressing model drift, and ensuring adaptation to changing data and tasks. For collaborative systems, these challenges are even more severe. Practitioners consider model deployment and monitoring phases both important and difficult. Models are often deployed as independent services, with limited adoption of MLOps principles. Reported issues include difficulties in designing infrastructure architectures for production deployment and integration problems with legacy applications. Many models in production are not monitored; key monitoring aspects include inputs, outputs, and decisions. Challenges also include a lack of monitoring practices, the need to create custom monitoring tools, and selecting appropriate metrics ([zimelewicz2024ml](#)). Acxiom, when using LLMs and LangChain for creating audience segmentation systems, faced challenges in debugging complex workflows. Observability achieved through LangSmith optimized token usage and effectively scaled their hybrid model deployment ([zenml2025llmops](#)). Continuous monitoring of LLMs and proactive detection of anomalies and adversarial behavior are crucial for ensuring their integrity. Integrating LLMs into CI/CD pipelines presents concerns related to computational costs, inaccuracies, error handling, bias, and issues related to development, deployment, maintenance, and ethics ([pahune2025transitioning](#)).

The increased number of components in collaborative systems means increased monitoring complexity and more potential failure points. Each model (LLM, SLM), the routing mechanism, and the communication links between them need to be monitored for performance, resource consumption, error rates, and drift. Correlating logs and metrics from different components to diagnose system-wide issues can be highly challenging. Monitoring and debugging tools specifically designed for this distributed, heterogeneous model environment are needed. Collaborative systems have more components (LLM, SLM, router) than single-model systems. Each component is a potential point of failure and needs individual monitoring for health, performance (latency, throughput), resource usage, and output quality. Additionally, the interactions between components need monitoring, such as data flow, API call success rates, and communication latency. When problems arise (e.g., overall response quality degrades), pinpointing which component or interaction is faulty can be complex, requiring sophisticated logging, tracing, and root cause analysis capabilities. This significantly increases deployment and operational complexity. ****Open Issues:**** Develop standardized deployment patterns and best practices for LLM-SLM collaborative systems; create comprehensive monitoring frameworks capable of tracking request flows and performance metrics across multiple models; design automated anomaly detection and fault diagnosis tools for collaborative systems; research how

制; 研究针对混合模型架构的新型攻击向量; 为多模型系统建立全面的安全审计框架。

6 未来趋势

6.1 更智能、更自适应的协作框架

大型语言模型 (LLMs) 与小型语言模型 (SLMs) 之间的协作, 正从预设的、静态的交互模式, 向能够动态学习、适应并优化其协作策略的智能框架演进。这一趋势的核心在于赋予协作系统更大的自主性, 使其能够根据任务特性、数据变化和環境反馈灵活调整, 而非仅仅依赖固定规则或人工配置。未来的协作框架将越来越多地利用先进的机器学习技术, 例如强化学习和元学习, 以实现动态策略调整和高效的自优化。

6.1.1 基于强化学习的动态协作策略

强化学习 (RL) 为训练智能体在复杂动态环境中做出最优决策提供了强大的范式。在 LLM-SLM 协作中, RL 可用于学习任务分配、资源管理和交互协议等策略, 以最大化系统性能、运行效率或其他预定义目标。

一项重要进展是细粒度的动态资源分配。CITER 框架 ([zheng2025citer](#)) 利用 RL 训练一个路由器, 在词元级别决定下一个词元应由 SLM 还是 LLM 生成。该路由器的优化目标同时考虑了预测质量和推理成本, 学习预测词元级别的重要性并权衡其决策对未来生成序列的影响, 从而在效率与准确性之间实现动态平衡。这种词元级别的路由策略比传统的查询级别路由精细得多, 使得系统能够在生成任务的“较简单”部分利用 SLM 的效率, 而在“关键”词元上依赖 LLM 的能力, 从而可能带来显著的效率提升。

RL 也被应用于优化特定任务场景下的协作策略。在检索增强生成 (RAG) 任务中, Collab-RAG 框架 ([xu2025collab](#)) 使用 SLM 来分解复杂查询, 并通过一个黑盒 LLM 提供的反馈信号来提升 SLM 的分解能力。这种类似于 RL 机制的迭代优化过程, 增强了 SLM 在协作式 RAG 流程中的作用。此外, 研究者提出了 ReMA 框架 ([wan2025rema](#)), 使用多智能体强化学习 (MARL) 引导模型进行元思考行为, 其中一个高层元认知智能体和一个低层推理智能体通过 RL 进行协作。这表明 RL 不仅可以优化任务执行, 还可以优化协作框架内的战略规划。另外, 逆强化学习 (UDRL) 已被用于训练 SLM 进行可控提示生成, 以优化输出属性 (如长度和相关性) ([lin2025efficient](#))。虽然主要关注 SLM 训练, 但通过 RL 控制生成属性的原理可以扩展到协作场景, 例如当 SLM 输出需要满足特定 LLM 输入要求时。另一项研究提出了一种协作模式, 其中 LLM 生成简洁的思维链 (CoT) 指令, 然后由一个在 RL 优化的高密度 CoT 数据上微调过的 SLM 将其扩展为完整回答 ([she2025hawkeye](#)), 这展示了 RL 在优化结构化协作推理工作流程中 SLM 角色方面的潜力。

这些发展共同揭示了一个重要趋势: RL 正在推动协作模式从“预编程”向“基于学习”转变。早期的 LLM-SLM 协作通常依赖于简单、基于规则的路由逻辑。然而, 如 CITER 等框架所示, RL 可以基于质量和成本等复杂因素优化词元级别的路由决策, 这比静态规则要动态和精细得多。像 SmallPlan 和 Collab-RAG 这样的工作进一步表明, RL 可以使协作策略适应特定任务 (例如路径规划、RAG), 甚至基于 LLM 反馈改进协作组件 (如 SLM 查询分解器)。

to effectively manage and update individual components in distributed model deployments.

5.4 Security, Privacy, and Ethical Issues

While offering potential advantages, LLM-SLM collaborative systems may also introduce or amplify risks related to security, privacy, and ethics. The flow of data between different models, the collaborative decision-making process, and the inherent flaws of the models themselves can all be sources of these problems.

5.4.1 Privacy Risks of Data Flow Between Different Models

When data, especially data containing sensitive user information, flows between local SLMs and cloud LLMs (or even between different LLMs), the risk of privacy leakage increases.

Using cloud API calls for LLMs is common due to resource constraints, which inherently raises privacy concerns. SLMs are often deployed on edge devices to protect privacy (wang2024comprehensive). LLMs might leak personally identifiable information (PII) from training data or interactions. Frameworks like CoGenesis aim to logically prevent privacy leaks by having SLMs process private data locally (zhang2024cogenesis). The CrossLM framework involves SLMs trained on private client data enhancing cloud LLMs, raising questions about knowledge transfer versus data leakage (deng2023mutual). Federated collaboration is considered a strategy for balancing privacy.

While cloud-edge collaboration distributes data processing, it also creates more nodes for data transfer or processing, potentially increasing the attack surface for privacy breaches if not carefully designed. LLM-SLM collaboration often involves data flow between models, possibly across trust boundaries (e.g., device-to-cloud). LLMs, if handling sensitive user data, pose privacy risks. On-device SLMs are often used to mitigate these risks by keeping sensitive data local. However, the collaboration itself might require sharing some information (intermediate results, context summaries) that could inadvertently leak private details. Ensuring that shared information is sufficiently anonymized or generalized while still being useful for collaboration is a significant challenge. This requires careful design of data flow protocols and potentially the adoption of privacy-enhancing technologies (PETs) within the collaborative framework. ****Open Issues:**** Develop privacy-preserving data sharing protocols for LLM-SLM collaboration; formalize privacy guarantees in hybrid systems; investigate the trade-offs between privacy preservation and collaborative performance; audit data flows to identify potential leakage points.

5.4.2 Potential Bias Amplification or Security Vulnerabilities in Collaborative Systems

Individual LLMs and SLMs may carry biases from their training data. When these models collaborate, these biases could be amplified, or new, emergent biases could arise due to interaction dynamics. Ensuring fairness and mitigating bias in the combined system is more complex than for individual models.

LLMs can perpetuate and amplify societal biases present in training data. SLMs, if trained on smaller or less diverse datasets, may also exhibit bias. Iterative learning among LLMs can amplify subtle biases (ren2024bias). The ViLBias framework notes that combining different modalities (text and image) can reveal biases not apparent in single modalities, suggesting model collaboration might have similar effects (raza2024vilbias). LLM routers themselves can be vulnerable; if an attacker controls routing to direct queries to biased models, it could be exploited to induce biased outcomes (lin2025life).

ReMA 框架将 RL 应用推向了更高层次，实现了分层的元认知与推理协作。这意味着 RL 不仅是在自动化决策过程，更是在教导系统如何有效地协作，这种学习由数据驱动，使系统能够适应任务的细微差别以及不同模型的优势与弱点。

6.1.2 基于元学习的动态协作策略

元学习，或称“学会学习”，旨在训练模型能够利用少量新数据快速适应新任务或新环境。在 LLM-SLM 协作中，元学习可用于开发自适应路由策略或协作机制，从而更好地泛化到不同的下游任务或变化的数据分布。

尽管在当前研究中，将元学习明确应用于 LLM-SLM 协作的案例仍在涌现，但已有若干方向显示出与元学习目标的强一致性。基于不确定性的 SLM 路由策略 (chuang2025confident)，即当 SLM 对其预测不确定时将查询传递给能力更强的 LLM，探索了路由策略对新数据集的有效泛化。他们提出了一个数据构建流程来生成与数据无关的保留集，从而能够在不需要大量新数据的情况下做出有效的路由决策。这种高效适应和泛化的目标正是元学习的核心。

这些探索表明，元学习有望解决协作框架中快速适应的挑战。协作系统需要在多样化的任务和领域中有效工作。为每个新任务重新训练复杂的强化学习策略成本高昂且效率低下。元学习通过在元训练阶段让学习算法接触一系列任务分布，来明确训练快速适应能力。因此，未来的研究可能会更深入地探索利用元学习来训练 LLM-SLM 系统中的“路由器”或“协调器”，使其能够仅用少量样本就为未见的新任务快速配置有效的协作策略。这不仅将显著降低 LLM-SLM 系统在新应用中的部署成本和时间，使其更具通用性，也意味着系统具备“学会如何学习协作”的潜力。

6.2 模型能力的深度融合

目前，大型语言模型（LLM）与小型语言模型（SLM）的协作正超越简单的任务路由或流水线处理，朝着能力深度融合的方向发展。未来的趋势不仅关注任务分配的优化，更致力于探索模型间更细粒度的交互，甚至可能深入到模型内部的神经网络层级。其目标是构建共享的理解或表示空间，从而实现能力的真正融合。

传统的 LLM-SLM 协作通常涉及 SLM 处理简单任务或充当初步过滤器，而 LLM 则处理复杂任务。然而，未来的协作模式将远超这种简单的层级划分，朝着更复杂、双向甚至迭代的交互方式演进。在这种模式下，不同的模型可能会相互咨询、验证输出或共同构建解决方案，形成一个动态的“模型社会”。

未来的协作将类似于一个模型的“社会”，其中每个模型扮演不同的角色并遵循多样化的交互协议，而非简单的层级结构（例如，LLM 作为领导者，SLM 作为助手）。简单的路由虽然高效，但可能无法充分利用每种模型类型的独特优势，或解决高度复杂的问题。引入诸如 LLM 增强 SLM 输入数据（例如 SynCID）或 SLM 验证 LLM 输出（例如 CoVer）等机制，可以创造更有价值的协作。可以预见，将出现更多样、更灵活的交互模式，例如迭代精炼、相互查询和集成方法，其中 SLM 和 LLM 从不同角度或针对解决方案的不同组成部分做出贡献。这将带来更稳健和更具创造性的问题解决能力，但也对模型间的编排和通信机制提出了更高的要求。

Bias can compound in collaborative systems. If an SLM routes a query based on a biased understanding of user demographics, and the selected LLM then generates content with its own biases, the final output could be doubly biased. The interaction itself might also create novel forms of bias not present in the individual models. Both LLMs and SLMs can be biased due to their training data or architecture. In a collaborative system, multiple models contribute to the final output or decision process. If different models have different biases, these can interact in unpredictable ways. A biased output from one model can become biased input for another, potentially amplifying the initial bias (e.g., a biased SLM routes to an LLM which then reinforces the bias). Mitigating bias in such systems requires evaluating not just individual components but the entire collaborative pipeline and its decision logic. ****Open Issues:**** Develop methods to detect and measure bias amplification in LLM-SLM collaborative systems; design fair routing and task allocation strategies; create techniques for debiasing outputs from multi-model systems; understand how different collaborative architectures affect bias propagation.

The complexity and distributed nature of LLM-SLM systems might introduce new security vulnerabilities. Attackers could target communication channels, routing mechanisms, or exploit weaknesses in one model to compromise the entire system. LLMs are susceptible to various attacks like prompt injection, jailbreaking, data poisoning, and PII leakage. Research shows LLM routers themselves have lifecycle vulnerabilities, including adversarial attacks and backdoors, with DNN-based routers being particularly vulnerable. If a router is compromised, it could maliciously route queries to inappropriate models or leak data. Interaction points between LLMs and SLMs (e.g., API calls, data exchange) become potential attack surfaces. If an SLM on an edge device is compromised, it could send malicious inputs to the cloud LLM, or vice versa.

Each additional model and communication link in a collaborative system potentially expands the attack surface. Vulnerabilities in the "weakest link" (a less secure SLM or router) in a collaborative system could compromise more powerful LLMs or the integrity/security of the entire system. LLM-SLM systems consist of multiple components (LLMs, SLMs, routers, communication channels). Each component and interface is a potential attack point. Routers could be attacked to force misrouting (sending all queries to an expensive LLM, or to a compromised/ineffective SLM). Communication channels, if insecure, allow data exchanged between models to be intercepted or tampered with. A compromised SLM could feed malicious data to an LLM (or vice versa), potentially leading to harmful outputs or system takeover. Securing a collaborative system requires a holistic approach, considering the security of each component and their interactions, which is more complex than securing a single monolithic model. ****Open Issues:**** Develop secure communication protocols for LLM-SLM collaboration; design robust and attack-resistant routing mechanisms; investigate novel attack vectors targeting hybrid model architectures; create comprehensive security auditing frameworks for multi-model systems.

6 Future Trends

6.1 Smarter and More Adaptive Collaboration Frameworks

The collaboration between Large Language Models (LLMs) and Small Language Models (SLMs) is evolving from preset, static interaction modes towards intelligent frameworks capable of dynamically learning, adapting, and optimizing their collaboration strategies. The core of this trend lies in endowing collaborative systems with greater autonomy, enabling them to adjust flexibly based on task characteristics, data changes, and environmental feedback, rather than relying solely on fixed rules or manual configuration. Future collaborative frameworks

更深度的融合探索了 LLM 和 SLM 在内部神经网络层级（而不仅仅是输入/输出层面）进行交互的可能性。这可能涉及共享激活值、相互影响注意力机制，或融合学习到的中间表示。这种方法旨在摆脱将每个模型视为独立黑盒的局限。

DeepMLF 模型 ([georgiou2025deepmlf](#)) 虽然是为多模态情感分析设计的，但其提供了一个富有启发性的“深度融合”概念。它引入了可学习的令牌，在预训练解码器的多个层级上整合多模态信息。这种跨层深度融合以及使用可学习令牌作为信息载体的概念，可以适用于 LLM-SLM 协作。SLM 可以处理特定类型的特征，然后将这些特征（例如激活值）注入到 LLM 的特定层中，反之亦然。这为模型间更紧密的耦合提供了路径，其中一个模型的内部处理可以直接影响另一个模型，从而可能产生更丰富的组合表示。另一项关于模型剪枝的研究 ([ding2025sliding](#)) 提出了一种“滑动层合并”方法，该方法基于预定义的相似度阈值动态选择和融合连续的层。虽然应用于单个 LLM 的剪枝，但基于相似度合并层的想法为融合来自 SLM 和 LLM 的对齐层（如果它们的表示空间能够对齐）提供了一种潜在的方法论。这些探索表明，大语言模型（LLM）与小语言模型（SLM）之间的界限可能会变得“更模糊”。黑盒式的输入/输出交互限制了模型间可交换信息的带宽和类型，因为内部表示（激活值、隐藏状态）通常包含丰富、细微的信息，而这些信息在最终输出中丢失了。DeepMLF 的成功突显了在单一模型内部进行深度、多层融合的优势，这一原则则可以扩展到模型间的协作。如果一个 SLM 专门提取特定的句法特征，那么这些特征（作为激活值）可以直接输入到处理同一文本以完成语义任务的 LLM 的相关层中，从而在更深层次上丰富 LLM 的输入。未来的研究可能会侧重于探索在特定层“嫁接”或“连接”SLM 和 LLM 的方法，或许会使用适配器模块或学习到的门控机制来控制信息流。这可能会催生出高效且强大的混合模型，其中 SLM 的优势（例如，专业化的特征提取）被深度整合到 LLM 更广泛的推理过程中。然而，这也带来了重大的架构和训练挑战，需要仔细对齐内部表示。

大型语言模型（LLMs）与小型语言模型（SLMs）之间的协作正从纯文本领域扩展到更复杂、更丰富的多模态理解（整合视觉、语音等）和具身智能（例如，与环境交互的机器人、虚拟智能体）领域。在这些新兴领域中，LLM-SLM 协作将在应对多样化的计算需求和专门的处理任务方面发挥关键作用。

6.2.1 整合视觉与语音等多模态信息的大模型与小模型协作

随着人工智能系统越来越需要处理和理解来自多种模态的信息，LLM-SLM 协作对于有效管理文本、图像、音频及其他数据类型不同计算需求和专门处理流程变得至关重要。多模态数据在数据量、特征提取和融合方面提出了独特的挑战。LLM-SLM 协作可以高效地分配这些任务，由 SLMs 处理高吞吐量、较低层级的模态处理，而 LLMs 则专注于高层级的跨模态推理与生成。

现有研究设想 LLM 智能体具备“大-小模型协作”能力，并处理可能涉及多模态输入的多轮对话，例如在搜索和推荐场景中 ([zhang2025survey](#))。一项研究描述了通过 LLM 蒸馏训练 SLMs，使其成为多模态上下文中的多任务学习者，特别是用于提示生成（例如，文本到图像、文本到模板）([lin2025efficient](#))，这表明 SLMs 可以在 LLM 的协助下专门处理多模态相关任务。VITA-1.5 模型 ([fu2025vita](#)) 是一个整合了视觉、语言和语音的多模态大语言模型

will increasingly leverage advanced machine learning techniques, such as reinforcement learning and meta-learning, to achieve dynamic policy adjustments and efficient self-optimization.

6.1.1 Dynamic Collaboration Strategies Based on Reinforcement Learning

Reinforcement Learning (RL) provides a powerful paradigm for training agents to make optimal decisions in complex dynamic environments. In LLM-SLM collaboration, RL can be used to learn policies for task allocation, resource management, and interaction protocols to maximize system performance, operational efficiency, or other predefined objectives.

A significant advancement is fine-grained dynamic resource allocation. The CITER framework ([zheng2025citer](#)) utilizes RL to train a router that decides at the token level whether the next token should be generated by the SLM or the LLM. The optimization objective of this router considers both prediction quality and inference cost, learning to predict token-level importance and weigh the impact of its decisions on the future generated sequence, thereby achieving a dynamic balance between efficiency and accuracy. This token-level routing strategy is far more granular than traditional query-level routing, allowing the system to leverage SLM efficiency for "simpler" parts of a generation task while relying on LLM power for "critical" tokens, potentially leading to significant efficiency gains.

RL is also being applied to optimize collaboration strategies in specific task scenarios. In Retrieval-Augmented Generation (RAG) tasks, the Collab-RAG framework ([xu2025collab](#)) employs an SLM to decompose complex queries and improves the SLM's decomposition ability through feedback signals provided by a black-box LLM. This iterative optimization process, akin to RL mechanisms, enhances the SLM's role in the collaborative RAG process. Furthermore, researchers have proposed the ReMA framework ([wan2025rema](#)), using Multi-Agent Reinforcement Learning (MARL) to guide models towards meta-thinking behavior, where a high-level meta-cognitive agent and a low-level reasoning agent collaborate through RL. This suggests RL can optimize not only task execution but also strategic planning within the collaborative framework. Additionally, Upside-Down Reinforcement Learning (UDRL) has been used to train SLMs for controllable prompt generation to optimize output attributes like length and relevance ([lin2025efficient](#)). While primarily focused on SLM training, the principle of controlling generation attributes via RL can be extended to collaborative scenarios, for instance, when SLM outputs need to meet specific LLM input requirements. Another study proposed a collaborative mode where an LLM generates concise Chain-of-Thought (CoT) instructions, which are then expanded into full responses by an SLM fine-tuned on RL-optimized high-density CoT data ([she2025hawkeye](#)), demonstrating RL's potential in optimizing the role of SLMs within structured collaborative reasoning workflows.

These developments collectively reveal an important trend: RL is driving the shift in collaboration modes from "pre-programmed" to "learning-based." Initial LLM-SLM collaborations often relied on simple, rule-based routing logic. However, as demonstrated by frameworks like CITER, RL can optimize token-level routing decisions based on complex factors like quality and cost, which is far more dynamic and granular than static rules. Works like SmallPlan and Collab-RAG further show that RL can adapt collaboration strategies to specific tasks (e.g., path planning, RAG) and even improve collaborative components (like SLM query decomposers) based on LLM feedback. The ReMA framework pushes RL application to a higher level, enabling hierarchical meta-cognitive and reasoning collaboration. This signifies that RL is not just automating decision processes but teaching the system how to collaborate effectively, driven by data, allowing it to adapt to task nuances and the strengths/weaknesses of different models.

(MLLM)。虽然它是一个单一的 MLLM，但其多阶段训练方法（逐步整合视觉和语音数据）以及模态冲突（语音数据可能降低视觉任务性能）等挑战，突显了 LLM-SLM 协作方法的潜在优势。专门的 SLMs 可以在将特征馈送到中心 LLM 之前，处理特定模态（如 VITA-1.5 中提到的语音编码器）的初始处理或编码，或者管理语音输出。Long-VITA 模型 ([shen2025longvita](#)) 专注于长上下文视觉语言理解。训练此类模型的复杂性和成本表明，SLMs 可能在处理部分视觉信息（例如，预处理、从片段中提取特征）或在由 LLM 编排的更大任务中管理较短上下文交互方面发挥潜在作用。

这些趋势指向一种“专用处理器”模式：在多模态场景中，SLMs 很可能充当特定模态的高效编码器/解码器或早期融合模块。视觉 SLM 处理视觉输入，音频 SLM 处理听觉输入，这些专门的 SLMs 将紧凑、信息丰富的表示馈送给核心 LLM，由后者执行跨模态理解、推理并生成多模态响应（或许在 SLM 解码器的帮助下）。这种架构有望实现更具可扩展性和成本效益的 MLLMs，从而实现更丰富的人机交互。它还允许独立更新或改进特定于模态的 SLMs，而无需重新训练整个 MLLM。

6.2.2 大模型-小模型协作在机器人及具身智能体中的应用

具身智能体（如机器人）在动态的现实世界环境中运行，需要具备实时感知、决策和行动能力。LLM-SLM 协作非常适合满足这些需求：LLM 提供高层推理与规划，而 SLM 则负责在资源受限的机器人硬件上进行底层控制、传感器数据处理和快速反应。

SmallPlan 框架直接针对这一点：一个 LLM 作为教师模型，训练轻量级 SLM 以完成机器人学中的高层路径规划任务。这些 SLM 通过 LLM 指导和模拟环境中的强化学习进行训练，能够为导航提供动作序列。这明确地将 LLM-SLM 协作应用于具身任务，展示了一种将 LLM 推理能力提炼为机器人可部署 SLM 的实用方法。一项研究 ([glocker2025llm](#)) 提出了一个由 LLM 驱动的具身智能体系统，用于机器人在家庭环境中自主管理物品。该系统集成了多个由任务特定 LLM（可能包括 SLM 或为专业化/更小规模而微调的 LLM）驱动的专门智能体（路由、任务规划、知识库）。该系统使用 RAG 进行记忆增强，并结合 Grounded SAM 和 LLaMa3.2-Vision 实现鲁棒的目标检测，展示了不同模型（例如，Qwen2.5 用于专门智能体，LLaMA3.1 用于路由）如何在复杂的具身任务中协作。此外，关于具身多模态大模型（EMLMs）的背景研究 ([chen2025exploring](#)) 强调了感知、语言和行动的整合，指出了可扩展性和泛化性等挑战，而 LLM-SLM 协作可能为此提供解决方案（SLM 用于机载感知，LLM 用于基于云的复杂规划）。

这些进展表明，分层控制和推理架构正成为具身人工智能的主导范式。机器人既需要高层的理解和规划（“清洁厨房”），也需要底层的、实时的感觉运动控制（“抓住海绵”、“避开障碍物”）。LLM 擅长常识推理、任务分解和理解自然语言指令，但对于直接的实时机器人控制而言，它们通常太慢或计算成本过高。相比之下，SLM 更适合在设备上部署，处理诸如局部感知、反射性动作或执行由 LLM 提供的明确定义的子目标等任务。未来的系统很可能由 LLM 负责解释用户目标、将其分解为可执行的步骤并监控进度，而一组 SLM（或一个通用的机载 SLM）则通过与物理世界交互并向 LLM 发送反馈来执行这些步骤。这种架构有望使机器人更加智能、适应性强且能力出众，能够在人类环境中执行复杂、长周期的任务，同时利用

6.1.2 Dynamic Collaboration Strategies Based on Meta-Learning

Meta-Learning, or "learning to learn," aims to train models capable of rapidly adapting to new tasks or environments using minimal new data. In LLM-SLM collaboration, meta-learning can be used to develop adaptive routing strategies or collaboration mechanisms that generalize better to different downstream tasks or changing data distributions.

Although explicit applications of meta-learning to LLM-SLM collaboration are still emerging in current research, several directions show strong alignment with meta-learning objectives. Uncertainty-based SLM routing strategies ([chuang2025confident](#)), where queries are passed to a more capable LLM when the SLM is uncertain about its prediction, explore effective generalization of routing policies to new datasets. They propose a data construction pipeline to generate a data-agnostic hold-out set, enabling effective routing decisions without requiring large amounts of new data. This goal of efficient adaptation and generalization is central to meta-learning.

These explorations suggest that meta-learning holds promise for addressing the challenge of rapid adaptation in collaborative frameworks. Collaborative systems need to work effectively across diverse tasks and domains. Retraining complex RL policies for each new task would be costly and inefficient. Meta-learning explicitly trains for rapid adaptation by exposing the learning algorithm to a distribution of tasks during meta-training. Therefore, future research will likely delve deeper into leveraging meta-learning to train the "router" or "coordinator" in LLM-SLM systems, enabling them to quickly configure effective collaboration strategies for unseen new tasks using only a few samples. This would not only significantly reduce the deployment cost and time for LLM-SLM systems in new applications, making them more versatile, but also implies the potential for systems to learn how to learn to collaborate.

6.2 Deep Fusion of Model Capabilities

Currently, the collaboration between Large Language Models (LLMs) and Small Language Models (SLMs) is moving beyond simple task routing or pipeline processing towards deeper integration of capabilities. Future trends focus not only on optimizing task allocation but also on exploring interactions between models at finer granularities, potentially even at the neural layer level within the models. The goal is to construct shared understanding or representation spaces, thereby achieving a true fusion of capabilities.

Traditional LLM-SLM collaboration often involves SLMs handling simple tasks or acting as preliminary filters, while LLMs tackle complex tasks. However, future collaboration modes will far exceed this simple hierarchical division, moving towards more complex, bidirectional, and even iterative interaction methods. In such modes, different models might consult each other, verify outputs, or co-construct solutions, forming a dynamic "society of models."

Future collaboration will resemble a "society" of models, where each model plays different roles and follows diverse interaction protocols, rather than a simple hierarchy (e.g., LLM as leader, SLM as helper). Simple routing, while efficient, may not fully leverage the unique strengths of each model type or solve highly complex problems. Introducing mechanisms like LLMs enhancing SLM input data (e.g., SynCID) or SLMs verifying LLM outputs (e.g., CoVer) can create more valuable collaborations. It is foreseeable that more diverse and flexible interaction patterns will emerge, such as iterative refinement, mutual querying, and ensemble methods, where SLMs and LLMs contribute from different perspectives or to different components of a solution. This will lead to more robust and creative problem-solving but also places higher demands on inter-model orchestration and communication mechanisms.

Deeper fusion explores the possibility of LLMs and SLMs interacting at the internal neural network layer level, not just at the input/output level. This could involve sharing activation values, mutual influence on attention mechanisms, or fusing learned intermediate representa-

强大的基于云的 LLM 以及保持实时响应性和自主性的机载 SLM。

7 结论

大型语言模型（LLMs）的快速发展带来了人工智能能力的显著提升。然而，其高昂的计算资源需求、部署成本以及潜在的延迟问题限制了其在更广泛场景中的应用。与此同时，小型语言模型（SLMs）凭借其高效、低耗和易于部署的特点，在特定任务和资源受限环境中展现出独特优势，尽管其能力和泛化性相对有限。如何有效结合 LLMs 和 SLMs 的优势，弥补各自的不足，已成为推动 AI 技术向更高效、更经济、更普适方向发展的重要研究课题。

本文系统性地回顾和探讨了大小语言模型之间的协作机制与架构。首先阐述了 LLMs 和 SLMs 的基本概念、特点以及各自的优势与局限。在此基础上，重点分析了当前主流的协作模式，包括流水线式、混合/路由式、辅助/增强式、知识蒸馏驱动式以及集成/融合式协作。并深入探讨了实现这些模式的关键技术，如任务分配与智能路由、模型间通信与接口设计、模型融合与结果集成、状态同步与上下文管理以及动态资源调度与优化。此外，结合实际端侧需求，本文探讨了协作机制在多种应用场景中的实用价值，包括实时低延迟推理、隐私敏感与本地数据处理、任务特定定制与个性化服务、离线或弱网络环境以及能源受限与绿色 AI 场景。

尽管 LLM-SLM 协作研究已取得显著进展并展现出巨大潜力，但仍面临诸多挑战。这些挑战主要集中在：协作效率与系统开销方面，任务路由决策和模型间通信可能引入额外的延迟与成本；模型间一致性与兼容性方面，涉及如何确保不同模型在输出风格、知识范围及事实准确性上的统一；协作系统的评估指标与基准测试方面，目前缺乏能够全面衡量协作效果、效率与成本的标准化测试平台与指标；以及安全、隐私与伦理方面，需要应对模型间数据流动带来的隐私风险，以及协作系统中潜在的偏见放大与安全漏洞。这些挑战指明了未来研究的方向。

7.1 研究意义与价值

大语言模型与小语言模型之间的协作机制研究具有深远的理论意义和广泛的应用价值。协作的核心价值在于能够在不显著牺牲性能的前提下，显著降低对计算资源的需求、减少推理延迟并降低运营成本。这使得原本因高资源门槛而无法获取的先进 AI 能力，得以部署在更多的边缘设备和嵌入式系统上，从而促进 AI 技术的普及化和民主化，惠及更广泛的用户和行业。

正如本综述第 4 章所展示的，LLM-SLM 协作能够有效满足边缘设备对实时响应、隐私保护、个性化、离线运行和低功耗等方面的苛刻要求。无论是可穿戴设备上的即时语音助手、保护个人隐私的本地知识库问答系统，还是工业场景中的快速异常检测，协作机制都为在资源受限环境中实现复杂的智能任务提供了可行的解决方案。

通过合理的任务分解和模块化设计，协作系统能够将复杂任务分配给具有不同专长的模型。这种模块化不仅有助于提高特定任务的处理精度和效率，也使系统调试、更新和维护变得更加灵活。可以针对性地升级或替换特定的 SLM 模块，而无需重新训练或调整整个庞大的 LLM。

随着对 AI 能耗问题的日益关注，LLM-SLM 协作通过优化资源利用和减少不必要的计算

tions. This approach aims to break away from treating each model as an independent black box.

The DeepMLF model ([georgiou2025deepmlf](#)), although designed for multimodal sentiment analysis, offers an inspiring "deep fusion" concept. It introduces learnable tokens that integrate multimodal information at multiple layers of a pre-trained decoder. This concept of cross-layer deep fusion and using learnable tokens as information carriers could be adapted for LLM-SLM collaboration. An SLM could process specific types of features, then inject these features (e.g., activation values) into specific layers of an LLM, and vice versa. This suggests pathways for tighter coupling between models, where one model's internal processing can directly influence another's, potentially yielding richer combined representations. Another study on model pruning ([ding2025sliding](#)) proposed a "sliding layer merge" method that dynamically selects and fuses consecutive layers based on a predefined similarity threshold. Although applied to pruning a single LLM, the idea of merging layers based on similarity offers a potential methodology for fusing aligned layers from SLMs and LLMs if their representation spaces can be aligned.

These explorations suggest that the boundaries between LLMs and SLMs might become "softer." Black-box input/output interactions limit the bandwidth and type of information exchangeable between models, as internal representations (activations, hidden states) often contain rich, nuanced information lost in the final output. DeepMLF's success highlights the benefits of deep, multi-layer fusion within a single model, a principle extensible to inter-model collaboration. If an SLM specializes in extracting specific syntactic features, these features (as activations) could be fed directly into relevant layers of an LLM processing the same text for a semantic task, thus enriching the LLM's input at a deeper level. Future research will likely focus on exploring ways to "graft" or "connect" SLMs and LLMs at specific layers, perhaps using adapter modules or learned gating mechanisms to control information flow. This could lead to efficient and powerful hybrid models where SLM strengths (e.g., specialized feature extraction) are deeply integrated into the LLM's broader reasoning process. However, this also presents significant architectural and training challenges, requiring careful alignment of internal representations.

6.3 Collaboration in Multimodal and Embodied Intelligence

The collaboration between Large Language Models (LLMs) and Small Language Models (SLMs) is expanding from the purely textual domain into more complex and richer areas of multimodal understanding (integrating vision, speech, etc.) and embodied intelligence (e.g., robots, virtual agents interacting with environments). In these emerging fields, LLM-SLM collaboration will play a crucial role in addressing diverse computational needs and specialized processing tasks.

6.3.1 Collaboration of Large and Small Models Integrating Multimodal Information like Vision and Speech

As AI systems increasingly need to process and understand information from multiple modalities, LLM-SLM collaboration becomes essential for effectively managing the different computational demands and specialized processing pipelines for text, images, audio, and other data types. Multimodal data presents unique challenges in terms of data volume, feature extraction, and fusion. LLM-SLM collaboration can efficiently allocate these tasks, with SLMs handling high-throughput, lower-level modality processing, while LLMs focus on high-level cross-modal reasoning and generation.

Existing research envisions LLM agents incorporating "large-small model collaboration" capabilities and handling multi-turn dialogues that may involve multimodal inputs, for ex-

销, 有助于构建更节能环保的"绿色 AI"系统。这对于 AI 技术的可持续发展至关重要。

总而言之, 大语言模型与小语言模型之间的协作, 不仅是应对当前 AI 技术挑战的有效策略, 也是激发 AI 应用创新、拓展 AI 能力边界的关键驱动力。

7.2 反思与展望

展望未来, 关于大语言模型与小语言模型之间协作机制的研究将继续朝着更智能、更深度融合、更广泛适用的方向深化。

目前许多协作策略仍依赖于预设规则或相对固定的任务分配。未来, 基于强化学习、元学习等先进技术的动态协作策略将成为主流。系统将能够根据实时任务复杂度、数据特征、可用资源乃至用户反馈, 自主学习和优化协作方式, 实现自配置与自适应。例如, CITER 框架中的令牌级动态路由, 以及通过强化学习优化 SLM 在 RAG 任务中的角色。这将使协作系统更加灵活高效, 更好地应对复杂多变的现实应用环境。

未来的协作将超越任务级分配或简单的输出组合, 探索模型间更深层的交互与能力融合。这可能涉及内部神经层级别的信息交换与影响, 例如共享激活值、注意力机制的相互引导, 甚至构建统一或对齐的表征空间。受 DeepMLF 模型、滑动层合并等技术启发, 打破模型黑箱, 实现内部状态的深度耦合, 有望催生出能力更强、协调更无缝的混合智能体。

随着 AI 技术向多模态理解和具身智能演进, LLM-SLM 协作将在处理和整合文本、视觉、语音乃至物理交互等多源信息流中扮演核心角色。在机器人应用中, LLM 可处理高层任务规划与理解, 而机器人硬件上的一系列专用 SLM 则负责实时传感器数据、执行底层控制指令并提供快速环境响应, 正如 SmallPlan 框架所展示的那样。这种分层协作将是构建能够在复杂物理世界中自主运行的智能体的关键。

随着协作系统能力的增强和广泛应用, 其决策过程的透明度、潜在安全风险(如对抗性攻击、隐私泄露)以及可能存在的偏见等问题将愈发突出。未来的研究需要投入更多精力开发有效的可解释性方法、构建鲁棒的安全防御体系, 并探索适用于协作智能系统的伦理准则与治理框架。

总之, 大语言模型与小语言模型之间的协作机制正处于快速发展阶段。它不仅为现有 AI 技术的瓶颈提供了创新解决方案, 也为人工智能的未来开辟了一片充满可能性的新大陆。我们有理由相信, 随着理论研究的深入和应用场景的不断拓展, 这种协作智能范式必将在未来的 AI 生态系统中扮演越来越重要的角色, 推动 AI 技术朝着更高效、更通用、更智能、更负责任的方向发展。

ample, in search and recommendation scenarios (**zhang2025survey**). One study describes training SLMs via LLM distillation to become multitask learners in multimodal contexts, particularly for prompt generation (e.g., text-to-image, text-to-template) (**lin2025efficient**), suggesting SLMs can specialize in multimodal-related tasks with LLM assistance. The VITA-1.5 model (**fu2025vita**) is a Multimodal Large Language Model (MLLM) integrating vision, language, and speech. Although a single MLLM, its multi-stage training approach (progressively integrating visual and speech data) and challenges like modality conflict (speech data potentially degrading visual task performance) highlight the potential benefits of an LLM-SLM collaborative approach. Specialized SLMs could handle initial processing or encoding of specific modalities (like the speech encoder mentioned in VITA-1.5) before feeding features to a central LLM, or manage speech output. The Long-VITA model (**shen2025longvita**) focuses on long-context vision-language understanding. The complexity and cost of training such models suggest potential roles for SLMs in handling parts of the visual information (e.g., preprocessing, extracting features from segments) or managing shorter-context interactions within a larger mission orchestrated by an LLM.

These trends point towards a "specialized processor" model: In multimodal scenarios, SLMs will likely act as efficient encoders/decoders for specific modalities or as early fusion modules. A vision SLM processes visual input, an audio SLM handles auditory input, and these specialized SLMs feed compact, informative representations to a core LLM that performs cross-modal understanding, reasoning, and generates multimodal responses (perhaps with the help of SLM decoders). This architecture promises more scalable and cost-effective MLLMs, enabling richer human-computer interactions. It also allows for potentially updating or improving modality-specific SLMs independently without retraining the entire MLLM.

6.3.2 Application of Large-Small Model Collaboration in Robots and Embodied Agents

Embodied agents (like robots) operate in dynamic real-world environments, requiring real-time perception, decision-making, and action capabilities. LLM-SLM collaboration is well-suited to meet these demands: LLMs provide high-level reasoning and planning, while SLMs handle low-level control, sensor data processing, and rapid reactions on resource-constrained robot hardware.

The SmallPlan framework directly addresses this: An LLM acts as a teacher model to train lightweight SLMs for high-level path planning tasks in robotics. These SLMs, trained via LLM guidance and RL in simulation, can provide action sequences for navigation. This explicitly applies LLM-SLM collaboration to embodied tasks, showing a practical way to distill LLM reasoning into deployable SLMs for robots. One study (**glocker2025llm**) presents an LLM-powered embodied agent system for autonomous item management by robots in home environments. It integrates multiple specialized agents (routing, task planning, knowledge base) driven by task-specific LLMs (which could include SLMs or LLMs fine-tuned for specialization/smaller size). The system uses RAG for memory augmentation and combines Grounded SAM and LLaMa3.2-Vision for robust object detection, demonstrating how different models (e.g., Qwen2.5 for specialized agents, LLaMA3.1 for routing) collaborate within a complex embodied task. Furthermore, background research on Embodied Multimodal Large Models (EMLMs) (**chen2025exploring**) highlights the integration of perception, language, and action, pointing out challenges like scalability and generalization where LLM-SLM collaboration could offer solutions (SLMs for onboard perception, LLMs for cloud-based complex planning).

These advancements point towards a hierarchical control and reasoning architecture becoming a dominant paradigm in embodied AI. Robots need both high-level understanding and planning ("clean the kitchen") and low-level, real-time sensorimotor control ("grasp the

sponge," "avoid the obstacle"). LLMs excel at commonsense reasoning, task decomposition, and understanding natural language instructions, but they are often too slow or computationally expensive for direct real-time robot control. SLMs, in contrast, are better suited for on-device deployment, handling tasks like local perception, reflexive actions, or executing well-defined sub-goals provided by the LLM. Future systems will likely see LLMs interpreting user goals, breaking them down into actionable steps, and monitoring progress, while a team of SLMs (or a versatile onboard SLM) executes these steps by interacting with the physical world and sending feedback to the LLM. This architecture promises to make robots more intelligent, adaptive, and capable, able to perform complex, long-horizon tasks in human environments, while leveraging powerful cloud-based LLMs alongside onboard SLMs that maintain real-time responsiveness and autonomy.

7 Conclusion

The rapid development of Large Language Models (LLMs) has brought significant improvements in artificial intelligence capabilities. However, their high computational resource requirements, deployment costs, and potential latency issues limit their application in broader scenarios. Meanwhile, Small Language Models (SLMs), with their efficiency, low consumption, and ease of deployment, exhibit unique advantages in specific tasks and resource-constrained environments, although their capabilities and generalization are relatively limited. Effectively combining the strengths of LLMs and SLMs to compensate for their respective shortcomings has become a key research topic driving AI technology towards greater efficiency, economy, and universality.

This survey systematically reviews and discusses the collaborative mechanisms and architectures between large and small language models. It first elaborates on the basic concepts, characteristics, and respective advantages and limitations of LLMs and SLMs. Based on this foundation, it focuses on analyzing current mainstream collaboration modes, including pipeline, hybrid/routing, auxiliary/enhancement, knowledge distillation-driven, and integration/fusion collaboration. It also delves into the key technologies for implementing these modes, such as task allocation and intelligent routing, inter-model communication and interface design, model fusion and result integration, state synchronization and context management, and dynamic resource scheduling and optimization. Furthermore, driven by actual on-device needs, this survey explores the practical value of collaborative mechanisms in various application scenarios, including real-time low-latency inference, privacy-sensitive and local data processing, task-specific customization and personalized services, offline or weak network environments, and energy-constrained and Green AI scenarios.

Although research on LLM-SLM collaboration has made significant progress and shown great potential, it still faces numerous challenges. These challenges primarily concentrate on: collaboration efficiency and system overhead, where task routing decisions and inter-model communication may introduce additional latency and costs; inter-model consistency and compatibility, concerning how to ensure uniformity in output style, knowledge scope, and factual accuracy across different models; evaluation metrics and benchmarking for collaborative systems, highlighting the lack of standardized testing platforms and metrics capable of comprehensively measuring collaborative effectiveness, efficiency, and cost; and security, privacy, and ethics, addressing the privacy risks associated with data flow between models, as well as potential bias amplification and security vulnerabilities in collaborative systems. These challenges indicate directions for future research.

7.1 Research Significance and Value

Research into the collaborative mechanisms between large and small language models holds profound theoretical significance and broad application value. The core value of collaboration lies in its ability to significantly reduce the demand for computational resources, decrease inference latency, and lower operational costs without substantially sacrificing performance. This enables advanced AI capabilities, previously inaccessible due to high resource barriers, to be deployed on more edge devices and embedded systems, thereby promoting the popularization and democratization of AI technology, benefiting a wider range of users and industries.

As demonstrated in Chapter 4 of this survey, LLM-SLM collaboration can effectively meet the demanding requirements of edge devices regarding real-time response, privacy protection, personalization, offline operation, and low power consumption. Whether it's instant voice assistants on wearables, local knowledge base Q&A systems protecting personal privacy, or rapid anomaly detection in industrial settings, collaborative mechanisms provide feasible solutions for achieving complex intelligent tasks in resource-constrained environments.

Through reasonable task decomposition and modular design, collaborative systems can assign complex tasks to models with different specialties. This modularity not only helps improve the processing accuracy and efficiency for specific tasks but also makes system debugging, updating, and maintenance more flexible. Specific SLM modules can be targeted for upgrade or replacement without needing to retrain or adjust the entire massive LLM.

With growing concern over AI energy consumption, LLM-SLM collaboration contributes to building more energy-efficient and environmentally friendly "Green AI" systems by optimizing resource utilization and reducing unnecessary computational overhead. This is crucial for the sustainable development of AI technology.

In summary, the collaboration between large and small language models is not only an effective strategy for addressing current AI technology challenges but also a key driving force for stimulating AI application innovation and expanding the boundaries of AI capabilities.

7.2 Reflections and Outlook

Looking ahead, research on the collaborative mechanisms between large and small language models will continue to deepen towards directions that are smarter, more integrated, and more broadly applicable.

Many current collaboration strategies still rely on preset rules or relatively fixed task allocations. In the future, dynamic collaboration strategies based on advanced techniques like reinforcement learning and meta-learning will become mainstream. Systems will be able to autonomously learn and optimize collaboration methods based on real-time task complexity, data characteristics, available resources, and even user feedback, achieving self-configuration and self-adaptation. Examples include the token-level dynamic routing in the CITER framework and optimizing the role of SLMs in RAG tasks through reinforcement learning. This will make collaborative systems more flexible and efficient, better able to cope with complex and changing real-world application environments.

Future collaboration will move beyond task-level allocation or simple output combination to explore deeper interactions and capability fusion between models. This might involve information exchange and influence at the internal neural layer level, such as sharing activation values, mutual guidance of attention mechanisms, or even constructing unified or aligned representation spaces. Inspired by techniques like the DeepMLF model and sliding layer merging, breaking the model black box and achieving deep coupling of internal states promises to yield hybrid intelligent agents with stronger capabilities and more seamless coordination.

As AI technology evolves towards multimodal understanding and embodied intelligence, LLM-SLM collaboration will play a central role in processing and integrating diverse infor-

mation streams, including text, vision, speech, and even physical interactions. In robotics applications, LLMs can handle high-level task planning and understanding, while a series of specialized SLMs on the robot hardware manage real-time sensor data, execute low-level control commands, and provide rapid environmental responses, as demonstrated by the SmallPlan framework. This hierarchical collaboration will be key to building intelligent agents capable of autonomously operating in complex physical worlds.

With the increasing capabilities and widespread application of collaborative systems, issues concerning the transparency of their decision-making processes, potential security risks (such as adversarial attacks and privacy leakage), and possible biases will become more prominent. Future research needs to invest more effort in developing effective interpretability methods, constructing robust security defense systems, and exploring ethical guidelines and governance frameworks suitable for collaborative intelligent systems.

In conclusion, the collaborative mechanism between large and small language models is in a phase of rapid development. It not only provides innovative solutions to the bottlenecks of existing AI technologies but also opens up a new continent full of possibilities for the future of artificial intelligence. We have reason to believe that as theoretical research deepens and application scenarios continue to expand, this collaborative intelligence paradigm will inevitably play an increasingly important role in the future AI ecosystem, driving AI technology towards greater efficiency, universality, intelligence, and responsibility.